Chapter 1

Helium Step Heat Experiment on Goethite in Hydrothermal Conditions Indicates High Retentivity at Surface Temperature and is Consistent with Multiple Diffusion Domain Systematics

1.1 Abstract

U-Th/He geochronology of Goethite (FeOOH) is a powerful method to quantitatively date weathering and erosion processes on Earth's surface; however, precise interpretation of goethite U-Th/He dates is challenging because uncertainty persists regarding the mineralogical and environmental conditions under which radiogenic helium is retained and how it may be lost over time. Standard vacuum step heating experiments have been successfully used on a variety of mineral phases to characterize helium migration, but the results of such experiments on goethite have been dominated by dehydration-induced helium loss, raising questions about their relevance to helium retention and loss in nature (Farley, H. B. Monteiro, et al., 2024). Observations that thermal decomposition of goethite is greatly slowed in the presence of high water pressure led us to develop a new method to conduct helium step heat experiments under hydrothermal conditions (up to 86 bars and 300°C) within standard experimental setups. Using this method, we report results from the first helium step heat experiment of proton-irradiated goethite under hydrothermal conditions. Irradiated goethite sample Win-06-01b of the Flinders Ranges, Southern Australia and distilled water were encapsulated in a pure silica glass ampoule and were subjected to multiple prograde and retrograde thermal cycles over a temperature range of 170°C to 250°C under water vapor saturated conditions. Step durations varied from 2 hours to over a week, with low-temperature (<120°C) permeation steps bracketing higher-temperature steps to account for residual helium within the ampoule. The results show apparent helium diffusivities approximately seven log units lower than those observed in vacuum, with apparent diffusivities following a consistent linear array on an Arrhenius plot for cumulative fractional 3He loss exceeding 7%. Delta plots generated using this reference array with a calculated activation energy of 157.1± 5.3kJ/mol are consistent with a volume-diffusion helium loss

mechanism with MDD (Multiple Diffusion Domain) systematics. These results show high helium retentivity in natural goethite under hydrothermal conditions that extrapolate to robust helium retention at surface conditions even over billion-year timescales. The method presented in this work can be systematically applied to experimentally investigate effects of crystallite size, chemical composition, and radiation damage to refine these results and to assess the applicability of existing vacuum step heat data to helium retention and loss in goethite in nature.

port language changes from manuscript: No three methods: One main method, then bounding with steady state assumption. Always measure permeabilities

1.2 Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024). It is uniquely suited to date weathering and ferruginization processes on Earth's surface, including laterite/duricrust formation and recrystallization (H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Allard et al., 2018), and ferruginization of organic matter or paleosols (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann et al., 2017).

As with any radiochronometer, including U-Th/He geochronology and thermochronometry, knowledge of the conditions under which the system is open to loss or gain of parent or daughter nuclides from the environment is essential to make meaningful geological interpretations based on measured ages. For goethite U-Th/He geochronology, step heats of irradiated samples are often used to reveal the degree of partial helium loss using the 4He/3He method (Shuster, P. M. Vasconcelos, et al., 2005), with reported losses always non-zero but relatively low, ranging from 2-30% (Shuster, P. M. Vasconcelos, et al., 2005; Heim et al., 2006; Hofmann et al., 2017; Deng et al., 2017). Although one study attempts to explain the variable loss principally as a result of variable radiation damage (Bassal et al., 2022), the effects of temperature, crystallite size and chemical composition on helium retentivity in goethite have not been experimentally determined and remain elusive (Farley, H. B. Monteiro, et al., 2024).

Laboratory experiments of helium migration under controlled conditions have proven extremely useful in understanding the mechanisms and rates of helium migration in geological materials and have produced sensible results that are predominantly explainable by thermally activated volume diffusion (Reiners and Farley, 1997; Farley, 2000; Farley, 2007; Cherniak and Watson, 2011; Farley, 2018), albeit variably complicated by the effects of radiation damage and annealing (Shuster, Flowers, et al., 2006; Guenthner et al., 2013). This approach has also been leveraged for goethite; however, some complications arise when applying this approach to goethite. Above surface temperatures, goethite decomposes to hematite and water, often along crystallite elongation directions (Goss, 1987). Farley et al. (2024) showed experimentally that helium and water release are highly correlated even at temperatures less than 200°C, suggesting that goethite decomposition into hematite is frequently the dominant process governing helium migration during vacuum step-heating experiments. The results of in-vacuum experiments therefore need not reflect helium migration in goethite in nature.

Still, existing helium step heat measurements of goethite in vacuum likely remain informative to an extent, as goethite dehydration kinetics are highly influenced by some of the same factors that could be responsible for variable helium retention in natural goethite, like variable grain size (Navrotsky et al., 2008; Guo and Barnard, 2011; Goss, 1987), crystallinity (Schwertmann et al., 1985) or substitution of aluminum or other cations within the goethite structure (Majzlan et al., 2003; Bassal et al., 2022). For example, Shuster et al. (2005) initially conducted vacuum step-heating experiments on two vitreous, polycrystalline goethite samples, in which uniformly distributed 3He was introduced by proton irradiation for comparison with radiogenic 4He that was

partially lost in nature. The results of Shuster, P. M. Vasconcelos, et al. (2005) could be well-fit with a model of two distinct diffusion domains or migration mechanisms with significant differences in helium retention under surface conditions, e.g. crystallite interiors and intergranular space. Although these results may have been impacted by dehydration in vacuum, certain aspects may remain applicable to helium loss in goethite in nature if helium loss in nature and the laboratory advance through the samples with similar geometries. One important point is that if laboratory experiments degas domains in order of their helium retentivity in nature, even if the rates of loss are affected by goethite dehydration, then a $^4He/^3He$ experiment in vacuum can still be used to measure helium loss (Shuster, P. M. Vasconcelos, et al., 2005; P. M. Vasconcelos et al., 2013).

An additional complexity is that natural goethite almost always occurs as aggregates of minute crystallites with cross-sections smaller than 1 µm in nature. Neglecting to account for multiple diffusion domains can lead to inaccurate determinations of activation energy (E_a) and frequency factor (D_o/a^2) for helium diffusion (Farley, 2018). Detailed step-heating with temperature cycling, as has been done for hematite (Farley and Flowers, 2012; Farley, 2018), is needed to characterize domain size and activation energy for polycrystalline samples.

Molecular simulations indicate that helium diffusion in perfectly crystalline goethite should be highly anisotropic, essentially reducing to 1-dimensional diffusion along the b-axis (typically the long axis of the polycrystalline aggregate) that would amount to near-instantaneous helium escape from goethite (Bassal et al., 2022), which is clearly disputed by observed (U–Th)/He dates as old as ~275 Ma (Waltenberg, 2012) and dates that are concordant with $^{40}Ar/^{39}Ar$ ages of other phases (P. M. Vasconcelos et al., 2013). However, the models presented by Bassal et al. (2022) suggest that crystallographic defects caused by radiation damage could block these channels, thereby reducing He loss. While these simulations are informative, they alone are insufficient for drawing conclusions about the mineralogical and environmental conditions needed for accurately dating the crystallization of goethite with the (U–Th)/He method and require experimental verification.

Until now, no studies have reported helium step-heating results for goethite under hydrothermal (high pH_2O) conditions, which could help stabilize goethite or slow its decomposition to illuminate the processes relevant to helium loss in natural settings. If experimental conditions can be controlled to slow the decomposition of goethite such that the rate of natural helium loss in goethite greatly exceeds the rate of goethite decomposition into hematite and water, it would open up an experimental approach to systematically assess how factors like crystallite size, temperature, radiation damage, and chemical composition could affect helium mobility and retentivity in nature, enabling estimates of helium retention in previously untested natural samples.

1.3 Experimental Methods

This study builds upon previous helium step heat studies of the dense colloform goethite sample Win-06-01b (occasionally referred to as Win0601b) collected within float at the Flinders Ranges in Southern Australia (Waltenberg, 2012; H. Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages: 180-135 Ma (Waltenberg, 2012). To produce a homogeneous distribution of 3He, Win-06-01b was irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster and Farley, 2005) and subsequently was subjected to helium step heat measurements in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024). In their step heat experiment of Win-06-01b, Farley, H. B. Monteiro, et al. (2024) additionally measured evolved water using capacitance manometry to show that helium and water release were tightly correlated for goethite step heats in vacuum. Although they used subtly different heating schedules, the results of Farley,

H. B. Monteiro, et al. (2024) clearly agree well with H. Monteiro (2017).

In this study, an aliquot of the same irradiated Win-06-01b goethite used by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) was encapsulated with MilliQ water within a silica glass tube. Silica glass was chosen over other laboratory glasses such as borosilicate because it is the most permeable glass type to helium (Norton, 1953; Altemose, 1961). To prepare the silica glass tube, 10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thinner pipettes using a natural gas and oxygen glass-welding torch and the pipette ends were fused closed. After the glass had cooled, 1.18 mg of irradiated Win-06-01b goethite blades less than 2mm in length were carefully transferred to the bottom of the tube in the presence of an ionizing fan. Less than a microliter of distilled water was transferred to the sealed pipette using a thin glass capillary, submerging the goethite, after which the tube was gently tapped to liberate the bubbles and ensure that the water and goethite were efficiently packed at the bottom of the pipette. To seal the pipette, the contents of the ampoule were flash-frozen in liquid nitrogen and transferred to a saltwater-ice bath at a temperature of -10°C. While the pipette end of the tube remained frozen in the saltwater-ice bath, the top was interfaced with a vacuum line and the tube was pumped to a pressure of <0.1 torr. The advantage of this approach was that water would remain solid within the tube, only slowly sublimating to vacuum while atmospheric gasses would be readily pumped away. While it was desirable to remove atmospheric gasses to reduce experimental complexity by removing potentially reactive gasses like oxygen, the primary motivation for pumping the tube to rough vacuum was to aid the sealing process. When the tube contents were under vacuum, fused glass would deform inwards and self-seal, greatly improving the ease and quality of the seal while requiring far less heat input from the torch. After an acceptably low pressure was attained, the tube was closed to the vacuum line and the saltwater-ice bath was quickly replaced with a liquid nitrogen bath, ensuring that no ice was melted. The liquid nitrogen bath was preferable during sealing as it more effectively protected the goethite and ice from the high temperatures of the glass-welding torch. A fine-flamed welding torch was used to seal the top of the ampoule, which dropped into the liquid nitrogen bath and was retrieved by passing the liquid nitrogen through a sieve. Fig. 1.1 shows the sealed ampoule after the completion of the step heats with all of the original water remaining, a testament to the quality of the ampoule's seal, which had sustained a massive pressure gradient of over 11 orders of magnitude across the ampoule walls, with an inner vapor pressure of about 3×10^4 torr (~40 bars: Haar et al. (1984)) against the ~10^{-7} torr ultra-high vacuum of the lamp cell while the ampoule was held at 250°C for over a week.



Figure 1.1: A view of the entire experimental ampoule after the completion of the step heat experiment before it was broken for the final step in the resistance furnace. At room temperature, the tube is approximately half-filled with gas, one-third filled with liquid water, and one-sixth filled with <2 mm long shards of irradiated Win-06-01b goethite. After the experiment, a slight reddish tint was visible on the thin edges of some goethite grains, which is likely a small amount of hematite. A gel-like substance was observed lining the inner walls of the tube, which is likely to be amorphous silica dissolved from the tube and redeposited by the thermally-cycled, high pressure liquid water phase.

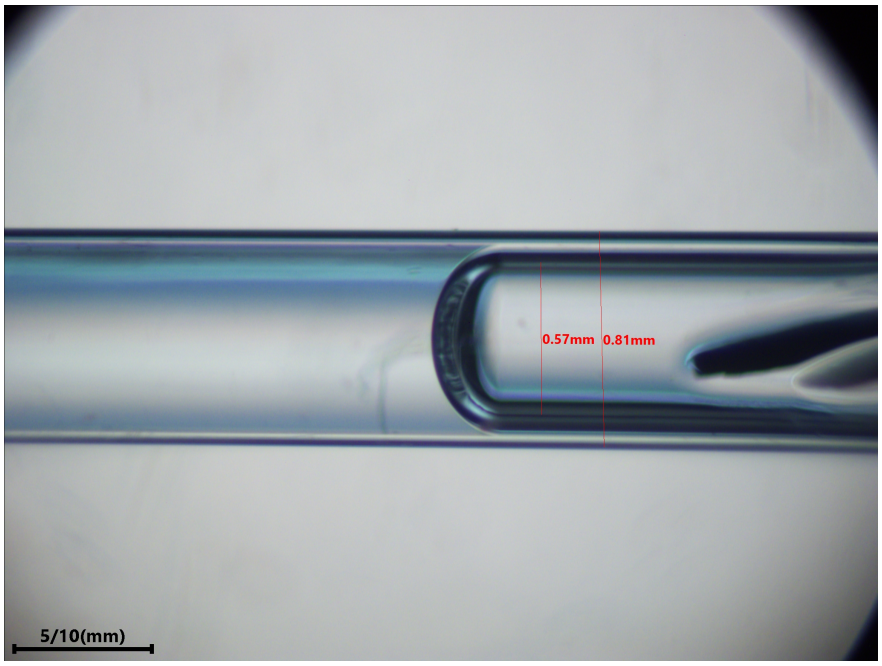


Figure 1.2: Measurements of the inner and outer diameter of the ampoule used in the hydrothermal step heats of Win-06-01b. The inner walls of the ampoule are only clearly visible within air bubbles.

The sealed ampoule, now referred to as HT-Win-06-01b, was placed in a copper foil packet and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley, Reiners, et al. (1999). Before the experiment began, the ampoule was loaded in the lamp cell, pumped to high vacuum, and was held at 70°C for 4 days to degas atmospheric helium entrained within the ampoule.

HT-Win-06-01b was subjected to multiple prograde-retrograde heating cycles to investigate possible MDD behavior. Step durations varied from 2 hours to about a week; the total experiment was conducted over about 40 days and included experimental steps ranging from 170-250°C. We elected not to heat the sample hotter because high temperatures generated very high vapor pressures. In early trials, we found that similar ampoules most often withstood the ~86 bar vapor pressures associated with liquid water saturation at 300°C (Haar et al., 1984), but they would explode when further heated to temperatures beyond 315°C.

Experimental temperature records show ~1°C amplitude oscillations around the target temperature. While we cannot exclude the possibility that sustained temperature gradients were present throughout the sample due to the use of the comparatively large ~1 mm thick ampoule, we are confident that the conclusions of this work are robust. The ampoule was in direct contact with the thermocouple junction and was in front of thermocouple junction with respect to the projector lamp heating source, suggesting that if there were a temperature discrepancy, the sample would've been hotter than the recorded temperature, which wouldn't affect the conclusions of this work. Additionally, during early trials of this method for helium step heats, silica-glass encapsulated Durango apatite had been subjected to step heats at Caltech and gave indistinguishable corrected 4He yields to unencapsulated Durango apatite, suggesting that the temperature recorded by the thermocouple for our experimental setup is sufficiently accurate.

Helium isotope measurements of evolved gasses from hydrothermal steps heats of glass-encapsulated irradiated HT-Win-06-01b goethite were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from January 31st, 2024 to March 13th, 2024. Helium was purified by exposing evolved gasses to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray, which was never significant during the experiment. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility.

A cold blank was measured at the beginning of the experiment, showing a negligible 3He blank essentially at 0 cps (technically this measurement yielded a slightly negative intercept to time 0) and a 4He blank signal corresponding to about 1.25 pccSTP, which was similar to 4He values attained during low temperature permeation step measurements. Permeation steps were thus assumed to measure shifting 4He blanks during the experiment, and the entire blank correction was added to all 4He uncertainties.

For the final step, the HT-Win-06-01b ampoule was broken and dried overnight in a 50°C oven, then completely degassed by heating to ~1100 °C in a resistance furnace. The final step measurement was conducted on a Thermo Fisher Helix SFT mass spectrometer. A hot furnace blank subjected to the same volume splits as the final HT-Win-06-01b measurement gave a 3He blank of

6.8 fccSTP and a 4He blank of 0.31 nccSTP. This blank correction was less than 1% of the measured 4He and negligible for 3He. A re-extract of HT-Win-06-01b immediately following the measurement yielded above-blank 3He and 4He signals, which were added to the total measurement, although both constituted less than 1% of the final step yield. Sensitivity was calculated by measuring a 3He-enriched air standard with known 3He and 4He contents immediately before the measurement.

Permeation Lag Correction Because HT-Win-06-01b was encapsulated in an imperfectly helium-transmissive silica glass ampoule, permeation lag was an expected complication in these measurements. To address this, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C) steps where the helium in goethite at high pH2O was effectively immobile at experimental timescales, following Thyer et al., 2026. For 3He, measured signals at these low temperature permeation steps were entirely controlled by the process of permeation through silica glass, and therefore could be used to estimate the amount of residual 3He within the ampoule between high-temperature experimental steps. However, for 4He, the measured signals at low temperature permeation steps were most often indistinguishable from the blank, complicating the application of the Naive Bracketing approach as discussed in Thyer et al., 2026 to correct for permeation lag in this dataset.

As a result, the permeation lag is instead addressed using the "Treatment as Uncertainty" approach described by Thyer et al., 2026. In this approach, an upper bound constraint for the helium content of the ampoule at the end of each high-temperature step was calculated using Eq. (**??**) (Chapter **??**).

The application of this equation requires knowledge of the lower-bound permeability of the system. The permeability of ampoules used in hydrothermal step heat experiments cannot reasonably be assumed to match literature values for anhydrous silica glass, and therefore the 3He permeability of the ampoule was determined using sequential isothermal permeation steps as described in the naive method outlined by Thyer et al., 2026.

10 series of 3-4 consecutive permeation steps held at a single temperature between 90-110°C for 120-180 minutes each with 26 minutes of cooling time before being inlet into the Helix-MC-Plus10k were identified as suitable for this exercise. The 3He measurements in these series were fit to Eq. (**??**) using a non-linear least squares optimization (Fig. 1.3). This is accomplished with the Levenberg-Marquardt algorithm implemented in the curve_fit function in the SciPy package of python (Virtanen et al., 2020), weighted by each measurement's own uncertainty. One additional series that otherwise qualified was not well fit by this model and was not used further for clarity, though its fit value of λ was within uncertainty of the other series' fit values for λ.

An initial guess of $\lambda \approx 6\text{x}10^{-6}$ sec^{-1} and $n_0 \approx \frac{m_i}{1-e^{-\lambda t_i}}$ (evaluated at the first measurement of a series) pccSTP was passed to the optimizer for each series. Uncertainties from the model fits were calculated using the covariance matrices generated for each of the model fits. Although the initial helium amount n_0 was a free parameter in the optimizations, the rate constant of permeation (λ) was the only parameter of interest for this exercise:

All measured apparent 3He permeabilities were equal to or in excess of the literature values for anhydrous silica glass (Fig. 1.4), similar to the observations of Thyer et al., 2026. 4He permeability itself could not be independently measured in this experiment due to blank-level signals during the low temperature permeation steps and is assumed to similarly exceed the literature values for anhydrous silica glass. To evaluate the uncertainty due to the permeation lag, we use the literature values as a conservative (lower bound) estimate of the true permeability. λt (where $\lambda = GRTK$) is estimated here using the 4He permeability K reported by Swets et al., 1961, while the 3He permeability is taken to be the product of the 4He permeability and the relative 3He/4He

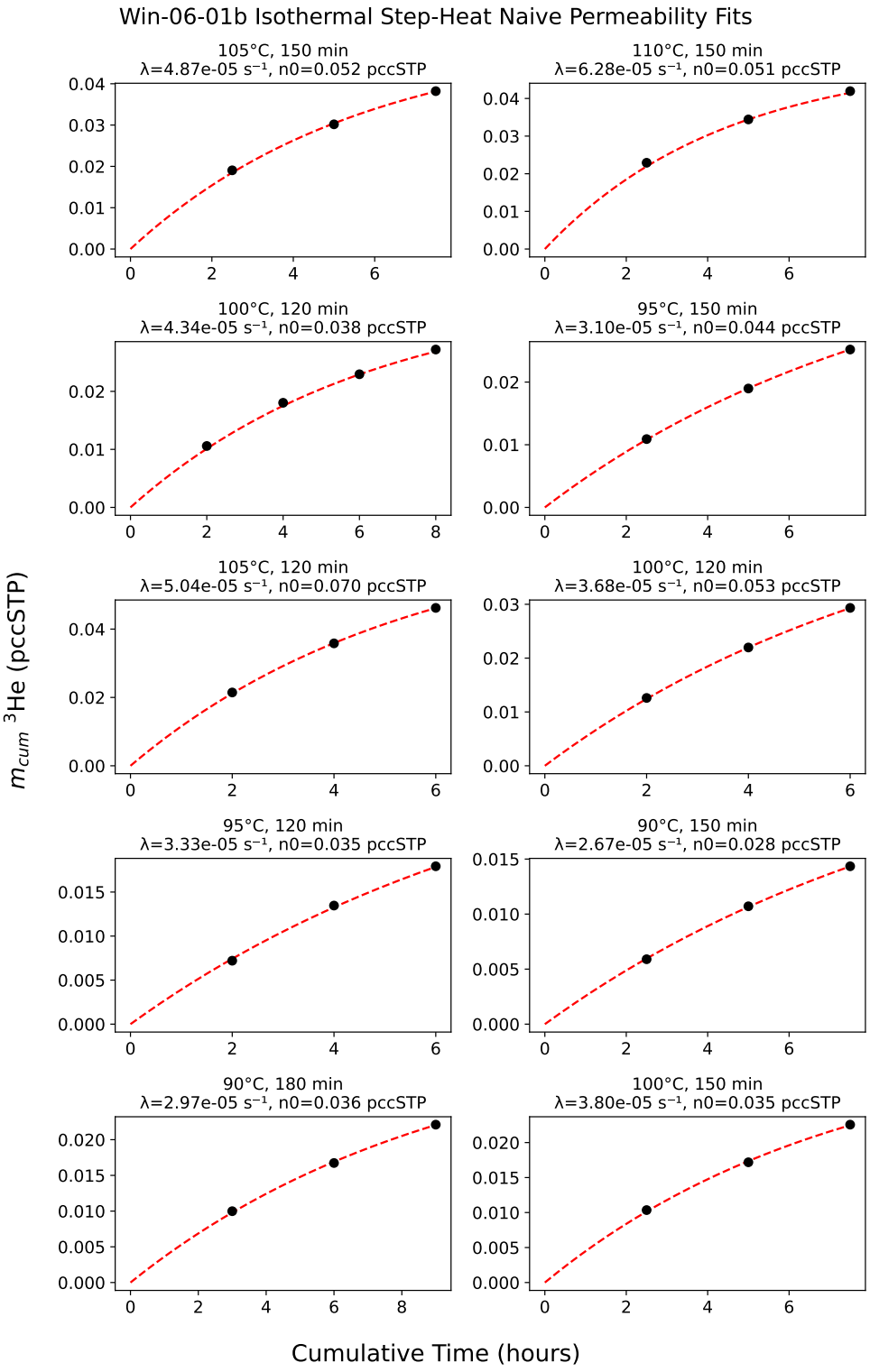


Figure 1.3: Cumulative yield against cumulative time for helium-3 yields for isothermal permeation steps. A non-linear least squares optimization fits λ and n_0 of Eq. (**??**) (Chapter **??**) to data from a series of permeation steps like these. The permeability K is calculated according to the relation $K = \lambda/GRT$ and is plotted as a function of temperature in Fig. 1.4.

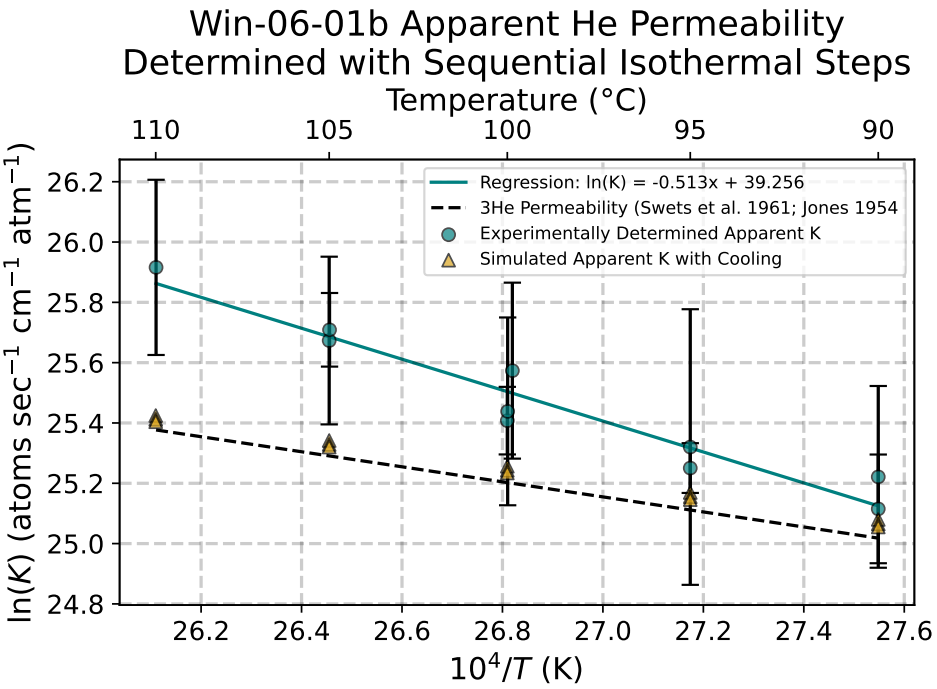


Figure 1.4: Arrhenius plot of apparent permeation coefficients as a function of inverse temperature for HT-Win-06-01b. The activation energy is higher than anhydrous silica glass, as expected (Masaryk, 1968; Shelby, 1972). It is unexpected that the glass ampoule of HT-Win-06-01b has higher apparent 3He permeability than the literature values from Swets et al., 1961 and Jones, 1953, but this can be explained by an increase in the partial pressure of the gas phase within the tube due to displacement by high He diffusivity, low He solubility water. Yellow points labeled "Diffusion Model with cooling, Anhydrous" were produced to evaluate the expected excess apparent permeability due to the 25 minute cooldown period before measurement. This is accomplished by simulating the measured apparent permeability for a dry silica glass ampoule that continued to permeate helium while cooling for an additional 26 minutes before measurement after being held at 90–110°C for 120–180 minutes, similarly to the measured values. Variability in these points is due to differences in the length of time the ampoule was heated but is very small. They show a near constant offset from the expected values of Swets et al., 1961 and Jones, 1953 that is much smaller than the uncertainty in the regression of the fit values determined experimentally in teal.

permeability reported by Jones, 1953. The geometric constant G is calculated using the measured inner and outer diameter of the ampoule (Fig. 1.2).

The first three experimental steps had values of λt lower than 4, and therefore the upper bound of the helium content of these steps could not be determined. These measurements are omitted from further analysis. Later experimental steps with values of λt lower than 4 could be determined using Eq. (**??**) from the supplement to Chapter **??** because upper bounds on previous steps' residual helium content could be evaluated.

1.4 Analytical Methods

Finite Difference Diffusion Model of Cooling Tube One potential concern about the results of the internally measured permeabilities is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder containing an arbitrary amount of helium gas within a helium-free chamber using the procedure described in (Thyer et al., 2026). The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at the end of each experimental step and a steady state concentration profile was assumed for the new arbitrary charge at T = 24°. Permeability values for 3He were assumed as the product of the 4He permeability values given by Swets et al. (1961) and the 3He/4He relative permeabilities given by Jones (1953).

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

Helium-in-Goethite Diffusivity Inversions Apparent values of D/a^2 for 3He and 4He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer (1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of the ampoules before and after each step, evaluated using (**??**). Diffusivity for step i is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional yields is largest, and minimized when the difference is smallest. This difference is maximized for the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag (${}^{i}_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$\begin{aligned} {}^{UB}D_\omega &= D\left(\Sigma^{i-1}F_m,\ \Sigma^i F_m + {}^{i}_{SS}n_t\right) \\ {}^{LB}D_\omega &= D\left(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^i F_m\right) \end{aligned} \tag{1.1}$$

For the 3He diffusivities, a linear regression was fit to the steps with cumulative 3He fractional yield ≥7% (12 steps, Fig. 1.7), yielding an R^2 statistic of 0.989. The fit's slope and intercept are

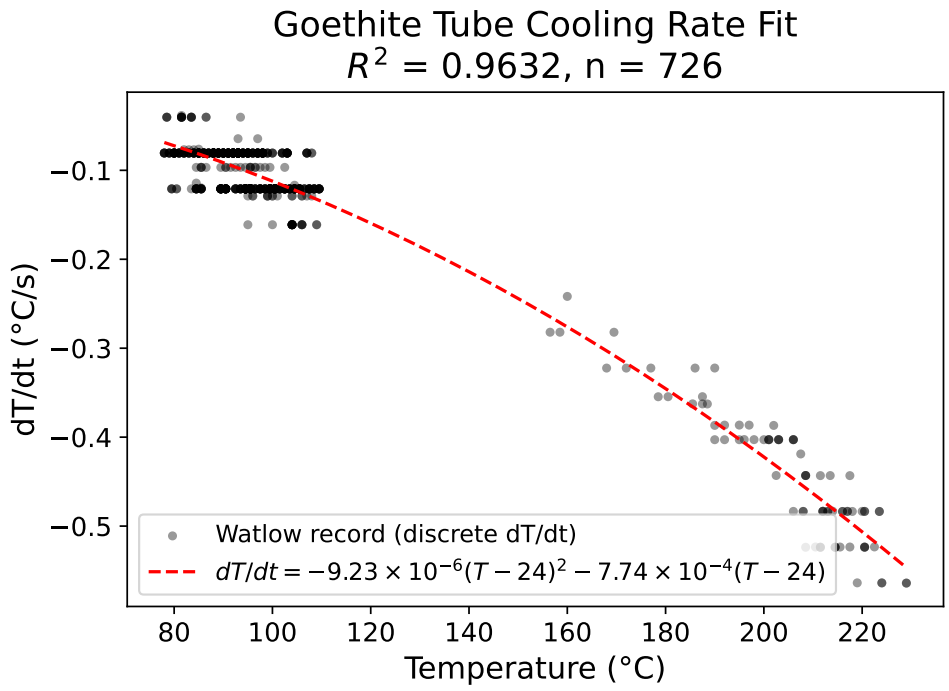


Figure 1.5: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

used as a reference line to calculate "Delta" values for the diffusivities of 3He and 4He (Fig. 1.8). No uncertainty on the total amounts of 3He and 4He is propagated into the diffusivities because the experiment operated at low cumulative yields where this error affects all diffusivities approximately homogeneously, so it is only relevant for the y-intercept of the regression and the corresponding calculated closure temperature. The uncertainty from the intercept and slope parameters of the linear regression is significantly larger than the uncertainty related to the total yield, so it is neglected.

Helium Isotope Ratio Uncertainty for Step Heats The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotopes systems:

1. $n_t = 0,\ n_0 = 0$
2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The 4He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal 3He permeation measurements show to underestimate the ampoule's true permeability (Fig. 1.4); 4He permeability itself could not be independently measured in this experiment (supplement, **??**). For the purposes of evaluating the measured isotope ratios, the relative 3He/4He permeability (δ) is taken to be 1.29, the 2σ upper bound of the value measured independently for a water-filled ampoule in Chapter **??** (Thyer et al., 2026): 1.10 ± 0.19 (2σ). 1.29 is in great excess of realistic values of the relative 3He/4He permeability, which is expected to be between 1.06-1.11 for the temperature range relevant to this experiment (Jones, 1953). These conservative assumptions taken together guarantee that the calculated upper and lower bounds for the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (**??**) (Chapter **??**): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $\frac{\omega_4}{\omega_3} = \frac{m_4}{m_3}$
2. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta \lambda t - 1}}$
3. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta \lambda t - 1} - \frac{m_3^{i-1}}{\delta (\lambda t)^{i-1} - 1}}$

4. $$\frac{\omega_4}{\omega_3} = \frac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1}-1}}{m_3 - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1}-1}}$$

Where m_x^{i-1} and $(\lambda t)^{i-1}$ refer to the measured helium content and λt of the previous step, respectively. λ here applies explicitly to 4He, whereas $\delta\lambda$ applies to 3He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

SEM Study of Thermally Treated Polycrystalline Goethite In early trials of the "glass tube method" for helium step heat experiments with controlled microenvironments presented here, sub-samples of goethite morphologically similar to Win-06-01b from Igarapé Bahia (Shuster, P. M. Vasconcelos, et al., 2005; H. Monteiro, 2017) were heated in vacuum using the projector lamp heating apparatus of Farley, Reiners, et al. (1999) and in sealed glass ampoules with gypsum or water additionally added to provide moisture to slow/prevent dehydration of goethite. All samples were held at 300°C for 4-11 hours, and the glass ampoules were broken and the goethites were allowed to air dry before all samples were placed on carbon tape and carbon coated. These samples were imaged at Caltech on Secondary Electron mode by the ZEISS 1550VP Field Emission SEM.

The resistance-furnace degassing step is detailed further in the Supplementary Information, in its **??** section. That same supplement also records, as a secondary and independent verification of the results and conclusions presented here, the naïve/informed bracketing method applied to this same dataset: an alternative treatment that explicitly tracks the ampoule's helium content step by step, using the isothermal-permeability fits established above, rather than bounding it as above.

1.5 Results

The complete corrected step-heat dataset underlying the diffusivities, delta plots, and Arrhenius regressions discussed below is distilled in Tables 1.1 and 1.2, reporting the lower-bound, central, and upper-bound values from the "treatment as uncertainty" permeation-lag correction for ^{3}He and ^{4}He respectively.

Table 1.1: ^{3}He step-heat dataset for HT-Win-06-01b under the "treatment as uncertainty" method: measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), the lower- and upper-bound true step production (ω_{LB}, ω_{UB}), and the corresponding lower-bound, central, and upper-bound apparent diffusivities. "–" denotes early steps with $\lambda t < 4$, for which bounds could not be determined (see 1.3). Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots (Fig. 1.21 and others derived from this dataset).

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	ω_{LB} (pccSTP)	ω_{UB} (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
170	300	0.9630	0.0000	1.1660	–	–	–	–	–
180	463	0.5442	0.9630	0.2413	–	–	–	–	–
190	550	0.5292	1.6241	0.1538	–	–	–	–	–
200	500	0.5190	2.2112	0.1430	0.3651	0.6619	-22.158	-21.775	-21.503
210	475	0.5770	2.7646	0.1440	0.4341	0.7210	-21.715	-21.406	-21.160
220	450	0.6333	3.3799	0.1447	0.4894	0.7781	-21.345	-21.068	-20.843
230	425	0.7899	4.0551	0.1685	0.6452	0.9584	-20.823	-20.604	-20.392
225	908	0.7513	4.8885	0.0716	0.5827	0.8229	-21.516	-21.246	-21.148
215	450	0.1415	5.6842	0.0349	0.0699	0.1764	-22.836	-22.124	-21.900
205	520	0.0763	5.8529	0.0184*	0.0413*	0.0947*	-23.479*	-22.863*	-22.646*
195	1106	0.0670	5.9491	0.0080*	0.0487*	0.0750*	-24.053*	-23.731*	-23.618*
185	740	0.0227	6.0246	0.0049*	0.0147*	0.0276*	-24.839*	-24.404*	-24.207*
175	920	0.0180	6.0610	0.0036*	0.0131*	0.0216*	-25.168*	-24.847*	-24.665*
200	260	0.0224	6.0814	0.0163*	0.0189*	0.0388*	-23.534*	-23.360*	-22.812*
210	520	0.0812	6.1081	0.0181*	0.0649*	0.0993*	-22.984*	-22.758*	-22.554*
220	1167	0.5766	6.1914	0.0447	0.5585	0.6213	-21.587	-21.553	-21.475
230	370	0.3577	6.7796	0.0903	0.3130	0.4480	-20.948	-20.811	-20.579
240	320	0.6767	7.1563	0.1776	0.5864	0.8543	-20.062	-19.913	-19.667
245	250	0.5549	7.8582	0.1888	0.3773	0.7437	-20.176	-19.778	-19.474
250	210	0.6934	8.4743	0.2786	0.5047	0.9721	-19.626	-19.296	-18.942
245	270	0.4659	9.2086	0.1413	0.1872	0.6071	-20.801	-19.874	-19.601
240	320	0.2857	9.7206	0.0749	0.1445	0.3606	-21.176	-20.487	-20.250
235	370	0.2199	10.0357	0.0517	0.1450	0.2716	-21.284	-20.864	-20.650
230	420	0.1400	10.2736	0.0303	0.0883	0.1703	-21.885	-21.421	-21.224
225	652	0.1494	10.4279	0.0206	0.1191	0.1700	-22.009	-21.781	-21.650
230	500	0.1550	10.5908	0.0271	0.1344	0.1820	-21.605	-21.462	-21.299
235	450	0.1951	10.7582	0.0361	0.1681	0.2313	-21.258	-21.107	-20.936
240	400	0.2425	10.9638	0.0483	0.2063	0.2908	-20.913	-20.750	-20.566

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Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	ω_{LB} (pccSTP)	ω_{UB} (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
245	350	0.3150	11.2328	0.0690	0.2667	0.3840	-20.495	-20.326	-20.125
250	300	0.3716	11.5689	0.0939	0.3026	0.4656	-20.181	-19.973	-19.743
250	300	0.3640	11.9627	0.0899	0.2700	0.4539	-20.261	-19.959	-19.734
250	300	0.3452	12.3490	0.0852	0.2553	0.4304	-20.285	-19.979	-19.755
250	10250	8.4829	12.7167	0.0494	8.3977	8.5324	-19.989	-19.976	-19.969
240	785	0.2261	21.2187	0.0209	0.1767	0.2470	-21.026	-20.778	-20.689
Total			146.1483						

Table 1.2: ^{4}He step-heat dataset and absolute ^{4}He/^{3}He ratios for HT-Win-06-01b under the "treatment as uncertainty" method, structured as in Table 1.1. "–" denotes steps for which bounds could not be determined. The Total row reports the bulk (total-yield) ^{4}He/^{3}He ratio, i.e. the normalizing value used to bulk-normalize these ratios for the plot in Fig. 1.21. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots, as in Table 1.1.

Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	$n_{t,\mathrm{UB}}$ (nccSTP)	ω_{LB} (nccSTP)	ω_{UB} (nccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{LB}}$	$(^4\mathrm{He}/^3\mathrm{He})_m$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{UB}}$
170	300	0.0192	0.0000	0.0280	–	–	–	–	–	–	–	–
180	463	0.0305	0.0192	0.0152	–	–	–	–	–	–	–	–
190	550	0.0558	0.0660	0.0179	–	–	–	–	–	–	–	–
200	500	0.1024	0.1375	0.0313	0.0845	0.1337	-26.207	-25.965	-25.619	–	197.3	–
210	475	0.1460	0.2500	0.0404	0.1147	0.1864	-25.314	-25.022	-24.717	248.2	253.0	268.8
220	450	0.2091	0.4055	0.0530	0.1686	0.2620	-24.407	-24.152	-23.876	326.3	330.1	349.3
230	425	0.3243	0.6319	0.0760	0.2714	0.4004	-23.425	-23.213	-22.956	403.7	410.6	432.3
225	908	0.3483	0.9799	0.0363	0.2722	0.3845	-23.807	-23.527	-23.413	442.6	463.6	474.4
215	450	0.0716	1.3487	0.0196	0.0353	0.0912	-24.945	-24.225	-23.976	429.1	505.9	537.1
205	520	0.0385	1.4359	0.0103*	0.0189*	0.0487*	-25.661*	-24.942*	-24.701*	392.4*	504.2*	534.6*
195	1106	0.0336	1.4912	0.0043*	0.0233*	0.0379*	-26.165*	-25.796*	-25.675*	446.7*	500.9*	515.3*
185	740	0.0079	1.5467	0.0019*	0.0037*	0.0098*	-27.586*	-26.808*	-26.592*	224.7*	350.1*	369.1*
175	920	0.0152	1.6020	0.0033*	0.0133*	0.0185*	-26.475*	-26.340*	-26.141*	843.2*	843.2*	982.0*
200	260	0.0107	1.6351	0.0091*	0.0073*	0.0198*	-25.787*	-25.411*	-24.791*	375.6*	475.5*	554.7*
210	520	0.2166	1.6547	0.0535*	0.2075*	0.2701*	-23.067*	-23.021*	-22.786*	2667.1*	2667.1*	3103.8*
220	1167	0.3187	1.8714	0.0269	0.2652	0.3456	-23.499	-23.302	-23.214	471.7	552.7	563.2
230	370	0.2113	2.6766	0.0590	0.1844	0.2703	-22.391	-22.249	-21.993	575.1	590.8	628.0
240	320	0.4286	3.1616	0.1243	0.3697	0.5529	-21.360	-21.203	-20.931	612.1	633.4	674.8
245	250	0.3988	3.6430	0.1521	0.2746	0.5510	-21.289	-20.899	-20.556	665.9	718.7	778.4
250	210	0.4440	4.2359	0.2020	0.2918	0.6460	-20.906	-20.469	-20.072	541.7	640.2	702.8
245	270	0.3123	4.7263	0.1053	0.1103	0.4176	-22.043	-20.981	-20.680	469.1	670.4	721.4
240	320	0.1927	5.0846	0.0559	0.0874	0.2485	-22.376	-21.575	-21.315	505.5	674.3	718.4
235	370	0.1517	5.3031	0.0393	0.0958	0.1910	-22.386	-21.922	-21.688	599.8	689.6	729.9
230	420	0.0991	5.4714	0.0236	0.0598	0.1227	-22.956	-22.448	-22.232	608.3	707.9	745.9
225	652	0.0995	5.5835	0.0151	0.0760	0.1146	-23.136	-22.863	-22.721	608.2	666.5	689.2
230	500	0.1051	5.6969	0.0202	0.0900	0.1253	-22.679	-22.523	-22.345	652.1	678.1	707.5
235	450	0.1369	5.8251	0.0279	0.1167	0.1648	-22.290	-22.128	-21.940	674.6	701.7	734.0
240	400	0.1781	5.9762	0.0390	0.1502	0.2171	-21.892	-21.719	-21.518	705.1	734.6	770.9
245	350	0.2305	6.1807	0.0556	0.1915	0.2861	-21.479	-21.290	-21.070	694.4	731.9	771.7

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Temp (°C)	Time (min)	^{4}He meas. (nccSTP)	Cum. ^{4}He (nccSTP)	$n_{t,\mathrm{UB}}$ (nccSTP)	ω_{LB} (nccSTP)	ω_{UB} (nccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{LB}}$	$(^4\mathrm{He}/^3\mathrm{He})_m$	$(^4\mathrm{He}/^3\mathrm{He})_{\mathrm{UB}}$
250	300	0.2777	6.4385	0.0758	0.2221	0.3535	-21.134	-20.906	-20.659	703.3	747.3	793.3
250	300	0.2654	6.7374	0.0725	0.1896	0.3379	-21.250	-20.908	-20.661	653.7	729.2	774.1
250	300	0.2489	7.0231	0.0680	0.1764	0.3169	-21.282	-20.932	-20.647	646.8	721.0	765.4
250	10250	6.8042	7.2901	0.0430	6.7362	6.8472	-20.704	-20.691	-20.682	800.6	802.1	803.3
240	785	0.1853	14.1156	0.0187	0.1423	0.2040	-21.688	-21.422	-21.325	770.5	819.4	838.0
Total			145.8145								997.7	

The diffusivities determined from the corrected step heat 3He and 4He yields for HT-Win-06-01b are always significantly lower than the diffusivities determined from step heat experiments on the same sample in vacuum, and evolutions that are distinct from those determined from the vacuum experiments. The Arrhenius plot of the 3He diffusivities initially follow a declining concave-up trend until ~3.5% cumulative 3He yield, then slowly decreasing in sub-parallel arrays during prograde and retrograde heating cycles from 3.5-7% cumulative 3He yields (Fig. 1.7, Fig. 1.8). The Arrhenius plot of the 4He diffusivities differs in that it is initially concave-down until a cumulative 4He yield of ~0.1%, after which it maintains a relatively constant linear array about one log unit lower than the 3He array at the end of the experiment (Fig. 1.7). The 3He diffusivities settled into a linear array which was suitable for linear regression after ~7% of the cumulative 3He yield was lost (Fig. 1.6). All remaining diffusivity measurements of the experiment were included in the linear regression, so that diffusivities from 3He loss fractions of about 7-15% were included. The linear regression yields an R^2 of 0.989, an activation energy of 157.1± 5.3kJ/mol, and a closure temperature of 98.1± 8.6°C assuming a cooling rate of 10°C/Ma and considering only the uncertainty in the regression intercept (D_0/a^2).

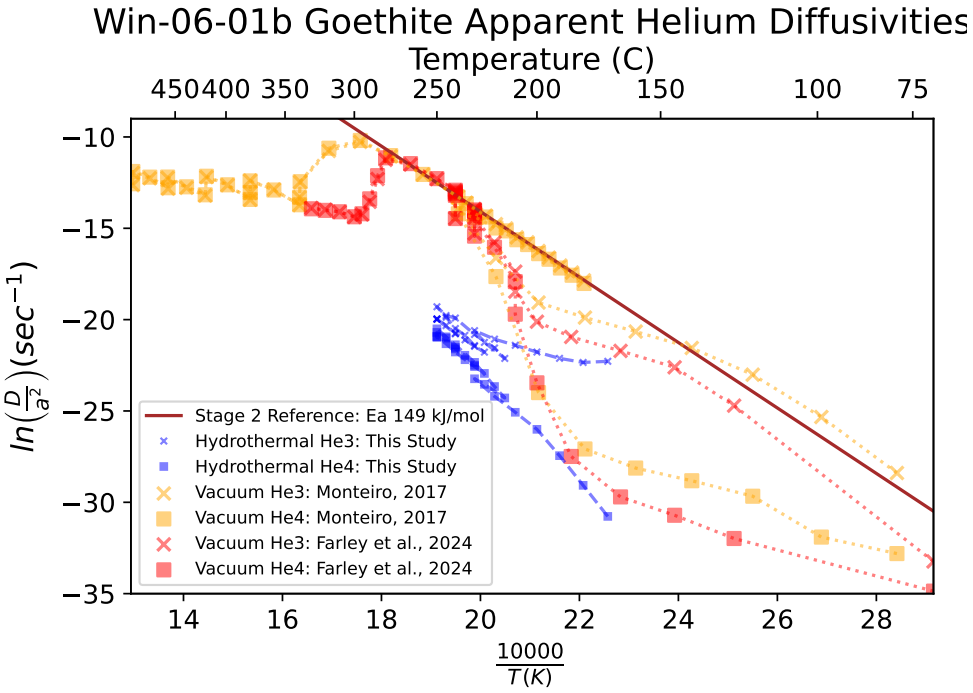


Figure 1.6: Arrhenius plot of 3He and 4He diffusivities from HT-Win-06-01b with other step heat experiments of Win-06-01b from H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024). The "Stage 2 reference line" is fit to helium step heats from H. Monteiro (2017). Error bars on the data from this study are not plotted for clarity. A diffusive domain corresponding to the Stage 2 reference line would have a closure temperature of about 45°C assuming a cooling rate of 10°C/Ma.

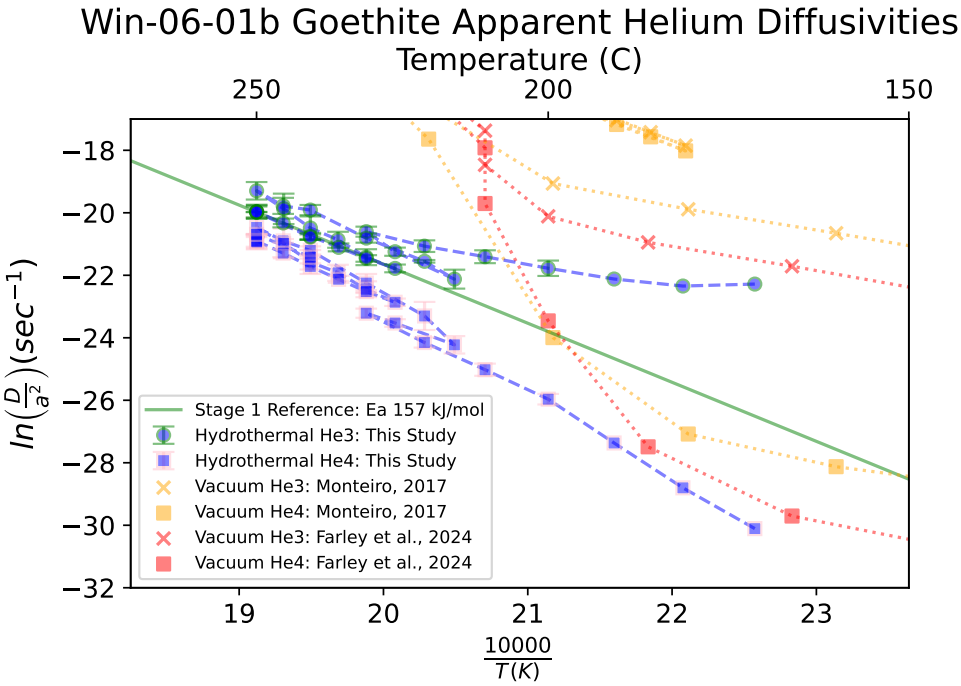


Figure 1.7: Linear regression of the last 11 measurements of 3He diffusivities in HT-Win-06-01b defining the "Stage 1" reference line. Apparent diffusivities under hydrothermal conditions are nearly 8 log units lower than in vacuum. A diffusion domain corresponding to the Stage 1 reference line has a closure temperature of 98.1±8.6°C assuming a cooling rate of 10°C/Ma. Apparent diffusivities calculated from 4He step heats are sometimes within error of those calculated for 3He, however they are occasionally resolved at the 2σ level and are about a single log unit lower than apparent diffusivities calculated for 3He after a cumulative fractional yield of 14% 3He.

A delta plot was generated using the intercept and slope parameters from the 3He regression as a reference line, similarly to previous work (Farley, 2018). The 3He apparent diffusivity delta values monotonically decrease until a cumulative 3He loss of 7%, after which the delta values are within error of Delta = 0: the reference line (Fig. 1.8). A series of low-yield measurements on the first retrograde cycle produce high-error diffusivities which appear as scattered, uncertain delta values just after a cumulative 3He loss of 4% for both 3He and 4He.

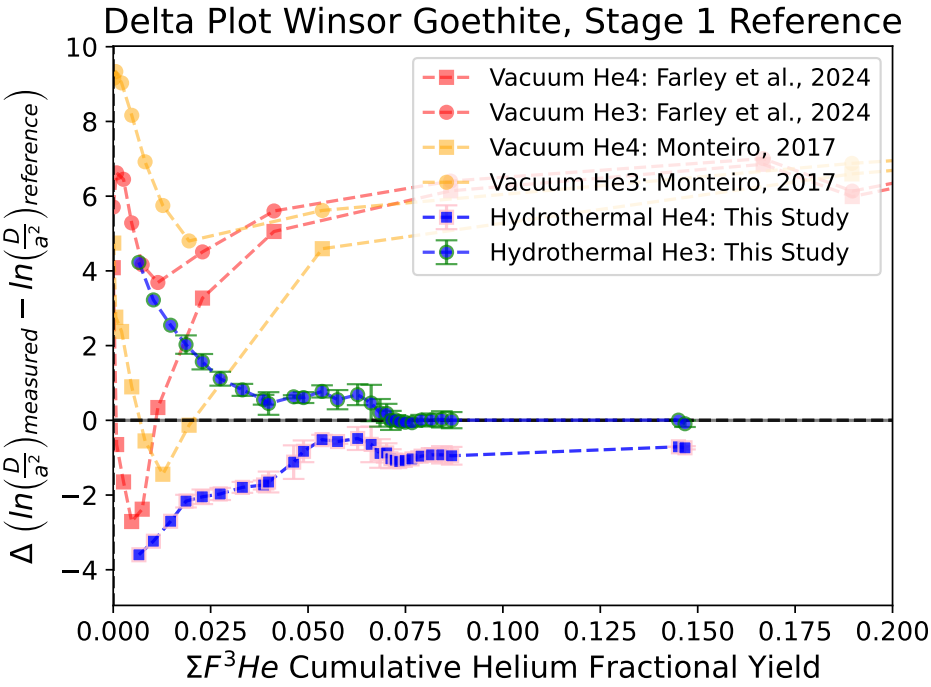


Figure 1.8: Delta plot similar to those presented in (Farley, 2018), or r/r0 plots in Lovera et al. (1991) and Farley and Flowers (2012). Delta plots are the difference between apparent log diffusivities and a reference line; they are designed to remove the temperature dependence of the dominant process of helium migration to reveal how apparent diffusivities evolve as an experiment proceeds. 3He monotonically decreases until settling on a temporary plateau from 4–6% 3He fractional yield, then settling on the reference line from 7–15% 3He fractional yield, which is consistent with MDD systematics. 4He diffusivities and delta values are generally highly uncertain but initially show an increase in delta values that settles to a slow increase or plateau about one log unit below the 3He reference line by a 3He fractional yield of 5%, which is consistent with diffusive loss of radiogenic 4He in the least retentive domains interrogated in HT-Win-06-01b. Apparent diffusivities for helium in Win-06-01b in hydrothermal conditions are up to 7 log units lower than the same sample in vacuum.

SEM imagery taken of thermally treated natural goethites in vacuum and open air showed the development of contraction cracks on the outside and throughout the interior of goethite grains heat treated in vacuum and in open air (Fig. 1.13 and Fig. 1.17), while goethite heat-treated when encapsulated with water or gypsum did not form cracks (Fig. 1.14 and Fig. 1.15). Goethite heat treated in open air turned red, whereas goethite heat treated in vacuum remained black.

1.6 Discussion

Measurement and Numerical Simulation of Apparent Permeability of Experimental Ampoule The apparent permeability values used in the numerical simulation (§1.4) are taken from the literature (Swets et al., 1961; Jones, 1953) and explicitly apply to an ampoule without water. However, the purpose of these simulations was simply to quantify the excess apparent permeabilities that would be observed for ampoules with a cooling history similar to that of the hydrothermal experiments. Because the predicted offset is universally small and a weak function of temperature or permeability, these simulations demonstrate that the much larger experimentally observed permeability is not an artifact of the cooling history of the hydrothermal experiments, reaffirming the conclusion that the hydrothermal experiments are indeed more permeable than the vacuum experiments.

The difference in log permeability between the actual and apparent permeabilities of the anhydrous simulation were significantly smaller than nearly all the 2σ uncertainties of the apparent permeabilities determined above for the hydrothermal Win-06-01b experiment, which ranged from 0.08 to 0.46 log units. Thus, no trends with step duration could be identified in the effective permeability values.

Therefore, the literature values underestimate the helium permeabilities of the ampoule used in this experiment, which is permissible because these values are used to calculate an upper bound constraint of the helium content of the ampoule, and lower permeabilities produce more conservative (higher) upper bound constraints.

The elevated measured permeability runs counter to expectations for a hydrothermal experiment: hydration of silica glass is generally reported to slow helium permeation at all temperatures (Tsuru et al., 2011), under the common paradigm that ions such as $(OH)^-$ clog interstitial space within the amorphous glassy network to slow permeation and decrease helium solubility (Norton, 1953; Shackelford, 1999). A similarly elevated permeability was independently observed at 100°C in a separate silica glass ampoule at Caltech partially filled with MilliQ water and equilibrated with ~0.1 torr of tank helium, as reported by Thyer et al. (2026), who posit that the elevated permeability is due to the low solubility and high diffusivity of helium in water, effectively shrinking the volume of the gas phase while maintaining equilibrium with the ampoule's interior throughout, leading to higher partial pressures of helium within the ampoule and therefore higher apparent permeabilities.

A linear regression of $\ln(K)$ against inverse temperature (Fig. 1.4) fit to these isothermal 3He permeation measurements over 90-110°C yields an R^2 of 0.936 and an activation energy of 10192±1882 (2 s.e.) cal/mol, a factor of two higher than expected for pure silica glass with or without water (Masaryk, 1968; Shelby, 1972), but similar to glasses with a high abundance of network-modifying cations such as Na, K, Pb, or Li (Norton, 1953; Altemose, 1961). Because 4He signals during these isothermal steps were blank-level (supplement, **??**), only the ampoule's 3He permeability could be measured directly this way, using the fitting procedure described above.

Consistency of the Bounding Method with Direct Permeation-Step Quantification: The Complete Degas Test All Win-06-01b step heat data presented in this chapter are reduced using the bounding method (Eq. (**??**)), because 4He permeation-step yields were near blank level and could not support the naïve/informed bracketing approach directly. As an internal check on this choice, the ampoule's 3He inventory was also reconstructed directly from the permeation steps themselves, using the isothermal rate constants of Fig. 1.4 (§**??**). Fig. 1.9 compares the two: n_t, the ampoule's residual helium quantified directly from a permeation step, against $n_{t,SS}$ (equivalent to $n_{t,\mathrm{UB}}$ used elsewhere in this chapter), the upper bound on that same residual calculated

independently from the preceding high-temperature step. For every ordinary permeation step in the experiment, $n_t \leq n_{t,SS}$, exactly as expected if the steady-state approximation genuinely bounds the ampoule's residual helium from above. This agreement is the justification for reducing the 4He dataset by the bounding method alone, since no analogous direct check of 4He is possible.

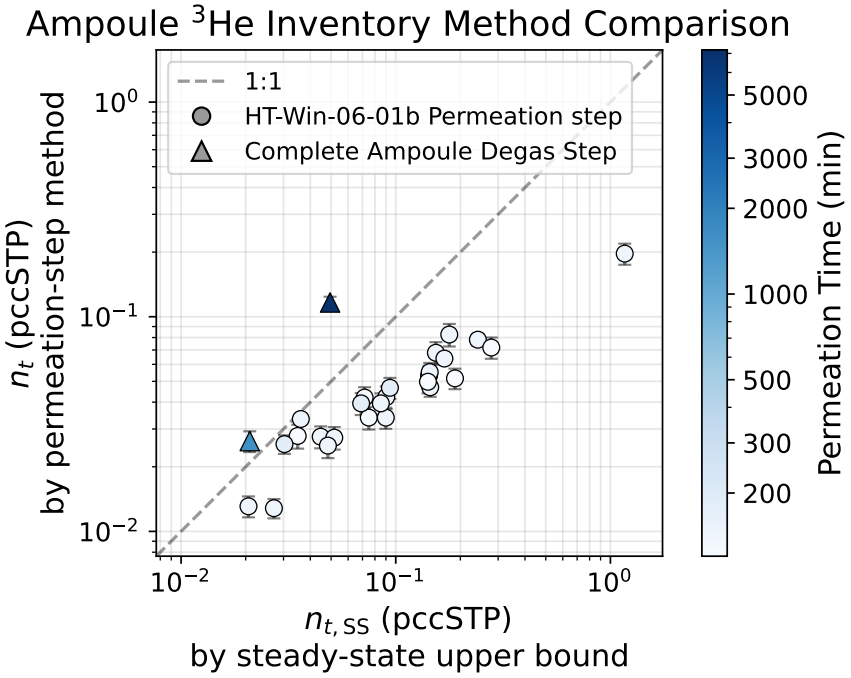


Figure 1.9: Two independent estimates of the ampoule's residual 3He inventory, n_t, following each permeation step of HT-Win-06-01b: quantified directly from the permeation step itself (*permeation-step method*, y-axis, §**??**), against the upper bound calculated independently from the preceding high-temperature step (*bounding method*, x-axis, Eq. (**??**)). Color indicates the duration of the permeation step. Circles are ordinary steps; triangles are the two "attempted complete degas" steps, held at 100°C for far longer than any other permeation step (1580 and 7200 minutes) in an attempt to drive the ampoule's 3He inventory toward zero. The dashed line is 1:1. Excepting the two complete-degas steps, n_t never exceeds $n_{t,SS}$, as required if permeation lag is genuinely bounded above by the steady-state approximation. Both complete-degas steps instead record more helium than their own $n_{t,SS}$ permits.

Beyond the ampoule's residual helium content itself, the two methods' derived apparent diffusivities agree just as closely. Fig. 1.10 compares $\ln(D/a^2)$ from the permeation-step method against the bounding method's central value for every step in the experiment, with the bounding method's own "treatment as uncertainty" spread on one axis and the permeation-step method's propagated uncertainty on the other; two steps are excluded from this comparison because their permeation-step n_0 depended on an atypically long, ~13-hour room-temperature permeation rather than the ~26-minute window used to calibrate every other step (§**??**). The two diffusivity datasets track the 1:1 line within uncertainty across the full range of the experiment. Fitting the permeation-step method's own diffusivities with the same Stage 1 procedure used for the bounding method above (Fig. 1.11) yields an activation energy of 157.0±4.8 kJ/mol (R^2=0.991, n=12) and a closure tem-

perature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (Fig. 1.7).

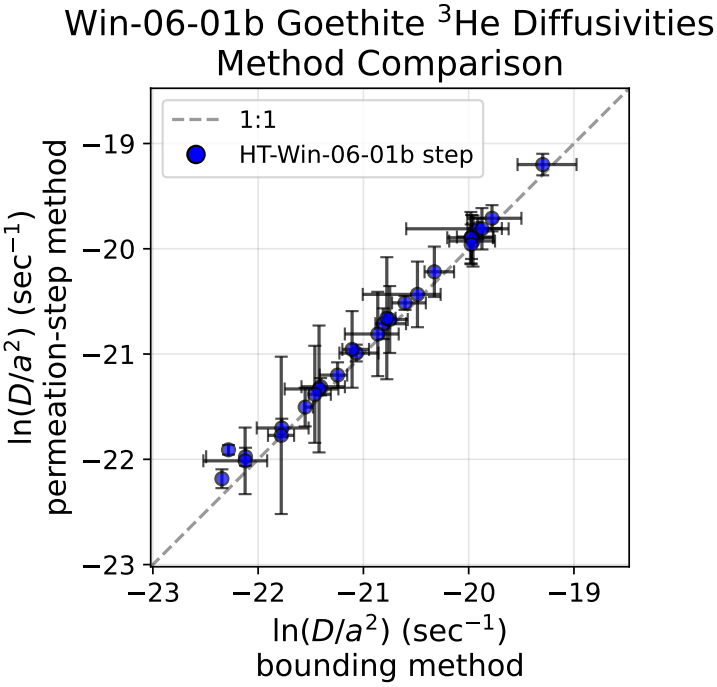


Figure 1.10: Apparent ^{3}He diffusivities for HT-Win-06-01b from the permeation-step method (y-axis, §**??**) against the bounding method's central value (x-axis, Eq. (**??**)), with the bounding method's "treatment as uncertainty" LB/UB spread and the permeation-step method's own propagated uncertainty plotted on their respective axes. The dashed line is 1:1. Two steps (195°C/1106 min and 185°C/740 min) are excluded for the reason given in §**??**.

The two "attempted complete degas" steps depart from this pattern. For the longer of the two (7200 minutes; predicted $n_t/n_0 \approx 1 \times 10^{-8}$ under simple first-order decay), the directly measured 3He inventory exceeds $n_{t,SS}$ by a factor of 1.5. For the shorter (1580 minutes; predicted $n_t/n_0 \approx 0.02$), the excess is smaller, 25% over $n_{t,SS}$. Because $n_{t,SS}$ is a conservative upper bound that assumes the sample's helium production rate is constant or decreasing during the step, a measured inventory in excess of $n_{t,SS}$ is not simply unlikely, it is inconsistent with that assumption, and requires the source term to have increased during the hold rather than decreased.

The size of the two excesses tracks the magnitude of the preceding, rather than the concurrent, thermal history. The 7200-minute step, with the larger (1.5×) excess, immediately followed a high-temperature step with a comparatively large 8.5 pccSTP 3He yield; the 1580-minute step, with the smaller (25%) excess, immediately followed a high-temperature step yielding only 0.2 pccSTP. An artifact intrinsic to the ampoule itself, such as an underestimated rate constant, a slow leak, or a memory effect in the extraction line, has no reason to scale with how much helium the preceding high-temperature step released from the goethite. That the excess does scale this way points to the sample rather than the ampoule: whatever reservoir is being interrogated by these extended low-temperature holds was charged in proportion to the immediately preceding high-temperature step, and released its contents slowly and with only weak temperature dependence, remaining measurable

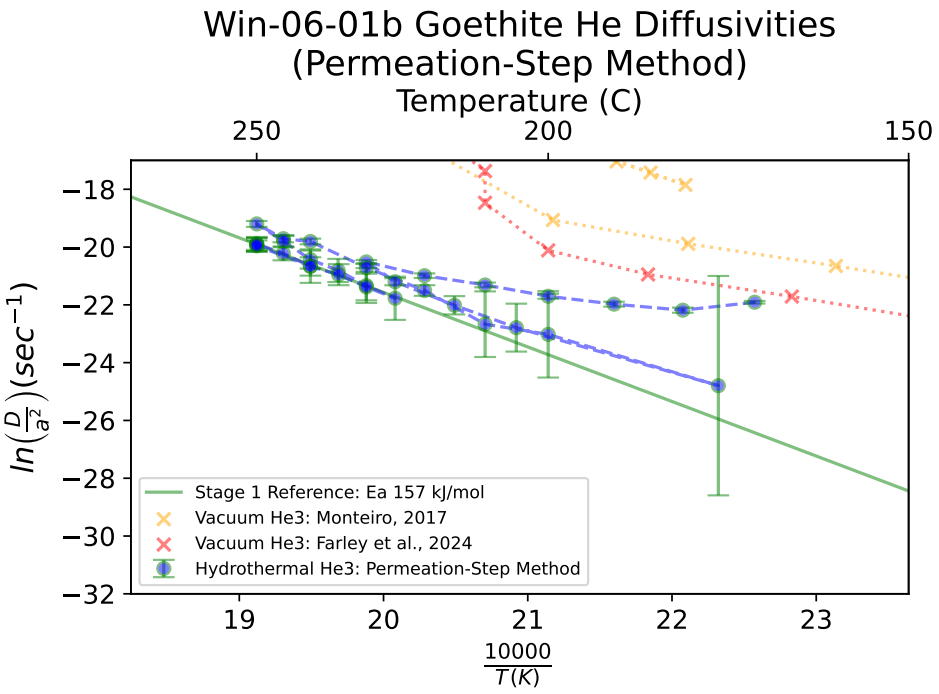


Figure 1.11: Analogue of Fig. 1.7 using the permeation-step method's own ^{3}He diffusivities and its own Stage 1 linear regression (§**??**), rather than the bounding method's. The resulting activation energy and closure temperature (157.0±4.8 kJ/mol; 97.5±7.8°C) agree closely with the bounding method's own (157.1±5.3 kJ/mol; 98.1±8.6°C, Fig. 1.7).

at 100°C over days to weeks.

Dehydration is not precluded by this pattern, but it is a harder case to make for these two steps specifically. Goethite dehydration in vacuum step heats is associated with anomalously elevated apparent diffusivity (§1.6; (Farley, H. B. Monteiro, et al., 2024)), yet the high-temperature steps immediately following each complete-degas hold, both at 250°C, return delta values indistinguishable from zero relative to the Stage 1 regression (Fig. 1.8) rather than the elevated delta expected if a fraction of the aggregate had newly dehydrated. The preferred explanation is instead diffusion of residual helium held within interstitial space of the goethite crystallite aggregate (§1.6; (Naono et al., 1987)), recharged during the preceding high-temperature step and released slowly, and with weak temperature dependence, during the following extended low-temperature hold. This requires communication between a lattice or grain-boundary source and an interstitial reservoir, a departure from the independent, non-communicating domains of a strict MDD interpretation, but it is consistent with the interstitial framework already invoked to explain the early, low-retention "Stage 1" fraction of 3He release (§1.6), and unlike dehydration, it requires no change in the intrinsic retentivity of the crystalline lattice recorded by the bulk of the experiment.

Consistency with MDD Systematics Previous work on helium step heats of goethite in vacuum have demonstrated degassing behavior in the laboratory that is inconsistent with thermally activated volume diffusion through goethite. For example, apparent diffusivities calculated from helium step heat experiments of goethites with uniform distributions of 3He produced by proton irradiation following the methods of Shuster, Farley, et al. (2004) will increase in an isothermal experiment (Farley, H. B. Monteiro, et al., 2024), which is inconsistent with volume diffusion through an unchanging crystal.

Farley, H. B. Monteiro, et al. (2024) reported that helium and water evolution in goethite vacuum step heating experiments are highly correlated and concluded that the crystal was decomposing into water and hematite during the experiment.

The goal of this work was to attempt to slow this decomposition process below the rate of thermal helium loss in unchanging goethite. SEM imagery shows that goethite that is thermally treated when encapsulated with water (or water for gypsum) does not extensively dehydrate like goethite heat-treated in vacuum or open air (Fig. 1.12–Fig. 1.17). While it is difficult to prove that decomposition or recrystallization is not playing a role in the results presented in this work, the apparent diffusivities observed for the Win-06-01b goethite under hydrothermal conditions are much lower than those observed in previous step heats of Win-06-01b in vacuum (Fig. 1.6, Fig. 1.7, Fig. 1.8, Fig. 1.18), implying that processes other than dehydration may be responsible for the observed apparent diffusivities.

All helium step heat data on goethites are from naturally occurring polycrystalline aggregates with unclear and potentially widely variable grain sizes (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024), which are expected to follow MDD systematics. If volume diffusion is responsible for the data reported here, they too would be expected to follow MDD systematics. In MDD systems, apparent bulk diffusivities of a homogeneously distributed diffusant at a single temperature will decrease as an experiment proceeds, as would delta or r/r0 values at any temperature (where $\Delta = ln\left(\frac{D}{a^2}\right)_{measured} - ln\left(\frac{D}{a^2}\right)_{reference}$) (Lovera et al., 1991; Farley, 2018), assuming that the activation energy of the reference line used in such plots matches that of the process controlling diffusion. For the activation energy of 160 kJ/mol retrieved by regression in this work, the homogeneously distributed 3He step heat data follow the expected MDD systematics and monotonically decrease within a delta plot (Fig. 1.8), which is consistent with volume diffusion in a MDD system.

The strong plateau in the delta plot and statistically robust regression over a wide range of

fractional cumulative 3He yield (from 7-15% ΣF_{3He} with an R^2 of 0.989) suggests that the frequency factor obtained by the regression may be dominated by a particular diffusion domain, rather than a mix of diffusion domains as is most often observed in the apparent diffusivities for polycrystalline hematite (Farley, 2018). The 4He apparent diffusivities are generally lower than the 3He apparent diffusivities and increased at the beginning of the experiment, which is consistent with rounding of 4He concentration profiles due to loss of radiogenic helium in the least retentive domains (Shuster, Farley, et al., 2004; Shuster, P. M. Vasconcelos, et al., 2005).

Many vacuum experiments in goethite also record declining delta values for early fractional yields of 3He (Shuster, P. M. Vasconcelos, et al., 2005; Farley, H. B. Monteiro, et al., 2024) that are generally not represented in the 4He data. This effect may be from helium migrating out of fast pathways along grain boundaries that would retain little helium in nature (Shuster, P. M. Vasconcelos, et al., 2005). Goethite aggregates are known to have interstices that are filled by chemisorbed non-stoichiometric water that represents the earliest water to be lost in vacuum (Naono et al., 1987). Non-stoichiometric hydrogen is known to persist in goethite even after heating to 140°C (Yapp and Poths, 1995). It is therefore reasonable to posit that artificial 3He observed in less-retentive domains may be trapped by non-stoichiometric water or hydroxyl groups within interstices of goethite aggregates. Dehydration of non-stoichiometric interstitial water may be associated with early fractions of degassing water and 3He from goethite samples during step heats, and could potential represent the LRD (low retention domain) of Shuster, P. M. Vasconcelos, et al. (2005), and the 3He offgassed during "Stage 1" in vacuum step heats (Farley, H. B. Monteiro, et al., 2024). The high variability of the portion of 3He hosted in early-degassing "Stage 1" fractions within vacuum step heats is sensible considering observations that goethite formed at different temperatures is known to have highly variable crystallinity, crystallite ordering, surface area, and non-stoichiometric water contents (Schwertmann et al., 1985). This population of 3He would likely also be the first to be lost under hydrothermal conditions.

While the results presented here are strong experimental evidence for high helium retentivity in goethite (Fig. 1.18, Fig. 1.19), they do not yet independently provide any evidence to support or refute hypotheses that helium diffusion in crystalline goethite is anisotropic or affected by radiation damage (Bassal et al., 2022).

Goethite Dehydration and/or Recrystallization During Helium Step Heat Experiments

The aim of helium step heat experiments is to characterize the mechanisms and rates helium loss in nature to aid in the interpretation of geochronologic and helium geochemical data. While the results of this experiment are encouraging in that they differ remarkably from experiments on the same and similar samples in vacuum that are primarily controlled by dehydration processes (Farley, H. B. Monteiro, et al., 2024), this experiment may also be affected to some degree by processes aside from volume diffusion that may potentially be unrelated to helium loss in goethite in nature.

Aside from nanoparticles (Guo and Barnard, 2011; Navrotsky et al., 2008), bulk goethite is unstable with respect to hematite and water at the pressures and temperatures of this experiment (Majzlan et al., 2003). The SEM study of goethite heat-treated in vacuum and open air shows that clearly visible volume contraction cracks form throughout the interiors and surfaces of highly-dehydrated goethite grains, while goethite heat-treated in water-saturated conditions does not (Figs. 1.13, 1.14, 1.15, 1.16, 1.17). Clearly, dehydration is occurring at a much smaller scale, if at all, in the hydrothermal experiments. While this evidence excludes the possibility of pervasive dehydration in the hydrothermal experiments, it does not exclude the possibility of some dehydration occurring at a scale too small to be observed in the SEM images. Future work on textural evolution of natural goethite aggregates during low fraction dehydration would be useful to resolve

the extent of "hidden" dehydration that may occur before the formation of SEM-visible cracks.

Dehydration processes may be responsible for some or all of the helium loss of Win-06-01b-HT, although if so they are markedly slowed relative to step heat experiments in vacuum (Fig. 1.19–Fig. 1.20). Even if dehydration completely controls helium loss in the step heat experiment of Win-06-01b-HT, these new data presented here would represent upper bounds for the apparent diffusivities of helium at the temperatures and fractional cumulative yields (ΣF_{He}).

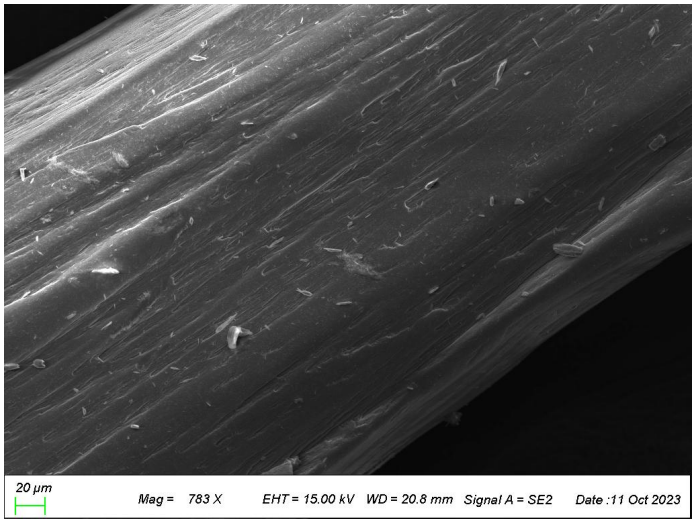


Figure 1.12: Secondary electron image of Bahia goethite aggregate before treatment.

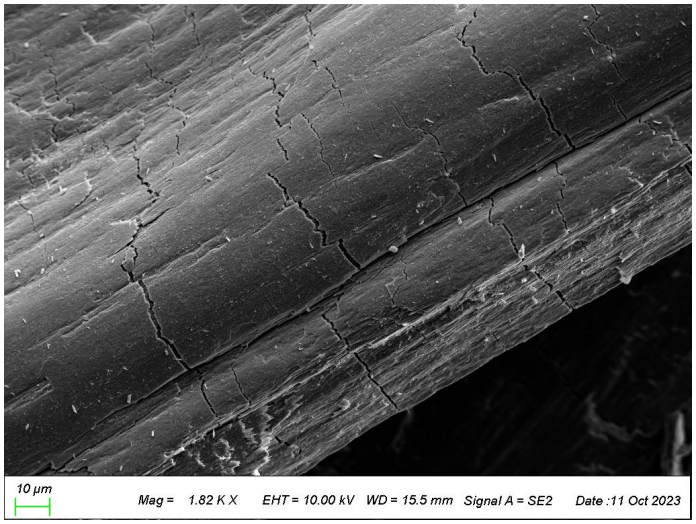


Figure 1.13: Secondary electron image of Bahia goethite aggregate heated at 300°C for 11 hours in vacuum. Cracks developed both parallel and perpendicular to the long axis of the polycrystalline aggregate, representing volume contraction due to dehydration/decomposition of goethite into hematite.

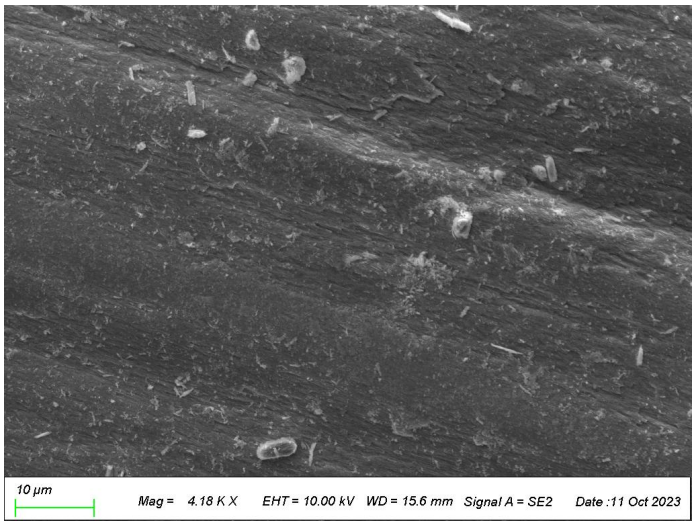


Figure 1.14: Secondary electron image of Bahia goethite aggregate heated at 300°C in a glass tube with gypsum. Despite treatment at high heat, no cracks indicative of dehydration were visible on the goethite surface, indicating that the gypsum dehydrated first and produced enough water pressure to stop/slow dehydration and prevent the development of microscopic cracks. At higher temperatures, goethite and gypsum reacted, forming a surface layer of other minerals.

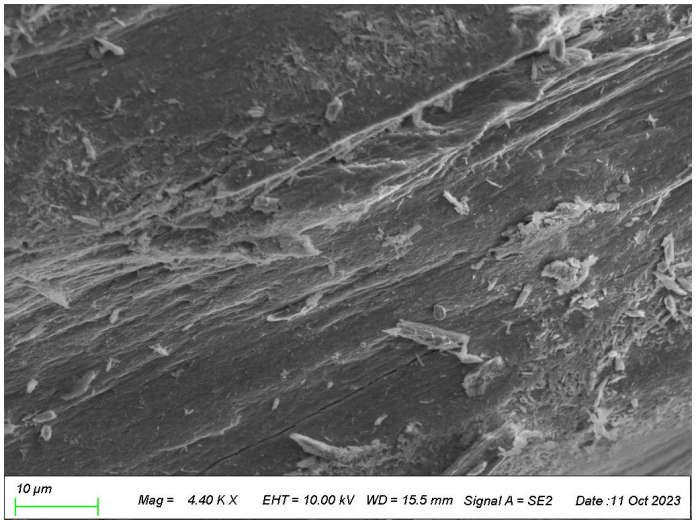


Figure 1.15: Secondary electron image of Bahia goethite heated at 300°C for 4 hours in a glass tube with water. No cracks were observed when investigating the goethite at similar scales to Figs. 1.12–1.13. Most of the aggregate's surface resembled Fig. 1.14. The area pictured had the roughest texture on the grain. A very small crack parallel to the long axis may indicate some amount of dehydration had occurred, or it could have been inherited from before hydrothermal treatment, since even the control untreated grain had some rough surfaces and rare long-axis-parallel cracks.

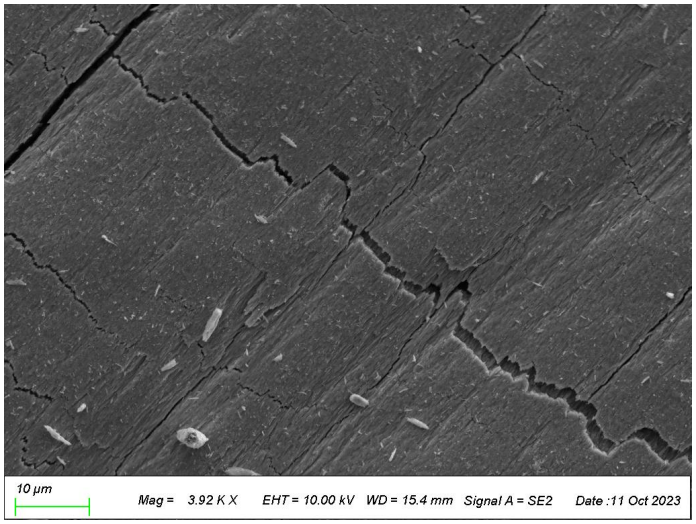


Figure 1.16: Secondary electron image of Bahia goethite heated at 300°C for 11 hours in vacuum, at the same scale as Fig. 1.15 and Fig. 1.14. This image shows rough cracks at various scales both parallel and orthogonal to the long axis of the goethite aggregate. These are typical decomposition features that would be expected in Fig. 1.15 and Fig. 1.14 if significant decomposition had occurred.

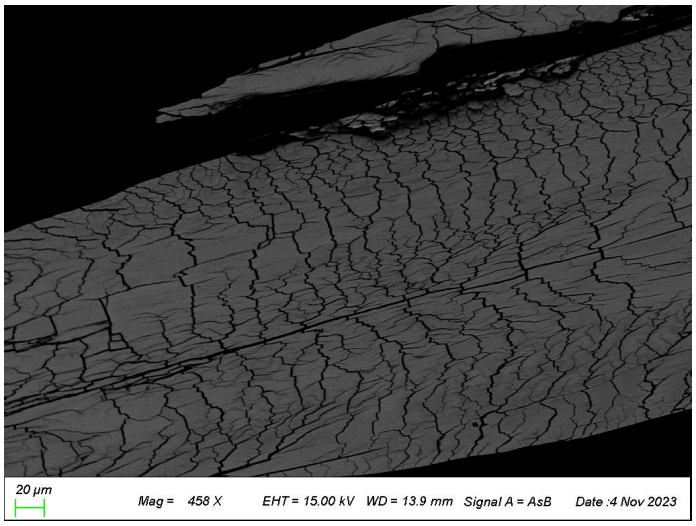


Figure 1.17: Backscatter electron image of a polished cross section of epoxy-mounted, vacuum-heated Bahia goethite (from the same subsample as Fig. 1.13) exposing the grain's interior. This image clearly shows that the cracks visible on the exterior of dehydrated goethite are pervasive throughout the interior as well.

The process of recrystallization is another potential complication in interpreting helium step heats of goethite in hydrothermal conditions. Poorly crystalline synthetic goethite held in hydrothermal conditions is known to recrystallize into more orderly, crystalline goethite aggregates, even at temperatures as low as 125°C in as little as 16 hours (Schwertmann et al., 1985). In this study, natural goethite was held in contact with liquid water and water vapor at hydrothermal conditions as harsh as ~40 bars and 250°C for as long as a week. These conditions appear to have visibly dissolved and reprecipitated silica from the ampoule's inner walls (Fig. 1.1), qualitatively testifying to the ability of the hydrothermal fluid to dissolve and recrystallize rock-forming oxides. It is unquestionable that recrystallization of some goethite could have occurred under these conditions; however, it is unclear if conditions were favorable to recrystallize this specimen of goethite, Win-06-01b, which is amongst the most helium-retentive and crystalline dated goethite samples (Waltenberg, 2012; Farley, H. B. Monteiro, et al., 2024). Less crystalline goethites in nature are commonly recrystallized, which appears to result in helium loss, resetting the U-Th/He age (H. S. Monteiro, P. M. P. Vasconcelos, et al., 2018; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Heim et al., 2006). Intuitively, it is difficult to imagine that dissolution-precipitation mediated recrystallization would not result in complete loss of helium within recrystallized regions. Therefore, assuming an initially uniform distribution of 3He throughout the crystal from proton irradiation, the upper bound extent of recrystallization would be equivalent to the 3He cumulative fractional yield, which is only 14.7% by the end of the hydrothermal experiment. Remaining helium is unlikely to pass through these now helium-free recrystallized areas if grain boundaries in goethite serve as fast pathways like they do in hematite (Farley, 2018). Recrystallization could be the dominant process controlling helium loss in this experiment, but only if it is faster than diffusive loss, in which case the data presented here would represent new upper bounds for apparent diffusivities for helium in Win-06-01b goethite. In that case, it stands to reason that processes such as dehydration and recrystallization may be faster than volume diffusion of helium in crystalline goethite in nature and may be the primary drivers of helium loss.

High Helium Retentivity of Win-06-01b-HT Apparent diffusivities calculated from helium step heats on the hydrothermal experiment on Win-06-01b goethite are clearly resolved and much lower than the apparent diffusivities calculated for the same sample in vacuum by H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) for all steps after the first few (Fig. 1.6, Fig. 1.7, Fig. 1.8). Diffusivities calculated from the hydrothermal experiment are ~8 log units lower than the earlier experiments in vacuum on the same sample. This high helium retentivity is explored through three lenses: closure temperature of cooling samples (Dodson, 1973), isothermal diffusion (Wolf et al., 1998), and through comparison with step heats of other polycrystalline samples. Strictly speaking, both closure temperatures and retention estimations under isothermal diffusion only apply to helium loss occurring exclusively via thermally activated volume diffusion with known domain size. While the results presented here on a polycrystalline sample are consistent with a thermally activated volume diffusive process for helium loss in goethite under hydrothermal conditions, they do not exclude other processes (such as goethite dehydration/decomposition/recrystallization) or potential complexities related to radiation damage and crystal defects on bulk diffusion (Bassal et al., 2022), nor do they permit estimation of the diffusivity at infinite temperature (D0).

If the slope and intercept on the regression of 3He apparent diffusivities from this experiment are taken to represent a real diffusion domain for goethite, then a closure temperature can be calculated for this domain. Considering uncertainty only in the intercept (D_0/a^2) term and a cooling rate of 10°C/Ma, the closure temperature for that domain would be 98.1± 8.6°C. This would correspond to the closure temperature of a ~0.2µm radius hematite crystal, or a ~650µm radius apatite crystal.

Goethite is formed near Earth's surface and thus U-Th/He geochronology is most typically applied to date surface processes (Heim et al., 2006; Yapp and Shuster, 2011; P. M. Vasconcelos et al., 2013; Danišík et al., 2013; H. S. Monteiro, P. M. Vasconcelos, et al., 2014; Hofmann et al., 2017; Allard et al., 2018). For crystals of goethite grown near the surface that are found near the surface, the concept of a closure temperature is poorly applicable, as it necessarily assumes a cooling rate (Dodson, 1973). In such cases, a better measure of retentivity is an analysis of isothermal retentivity over time (Wolf et al., 1998), which is commonly expressed in terms of t'/t, where t' is the helium age and t is the isothermal holding time (most often the 'real age'). For an isothermal holding time of 100 Ma at 40°C, a diffusive domain with the parameters obtained from the regression of 3He apparent diffusivities of Win-06-01b-HT has a t'/t = 0.968, where t'/t = 1 corresponds to perfect retentivity. Fig. 1.18 shows isothermal holding time against helium age for this same domain for 4 different temperatures. At 25°C, this domain is nearly quantitatively retentive of helium even at billion-year time scales. The results presented here qualitatively support that goethite is highly retentive of helium at surface conditions, even at 100-million-year timescales.

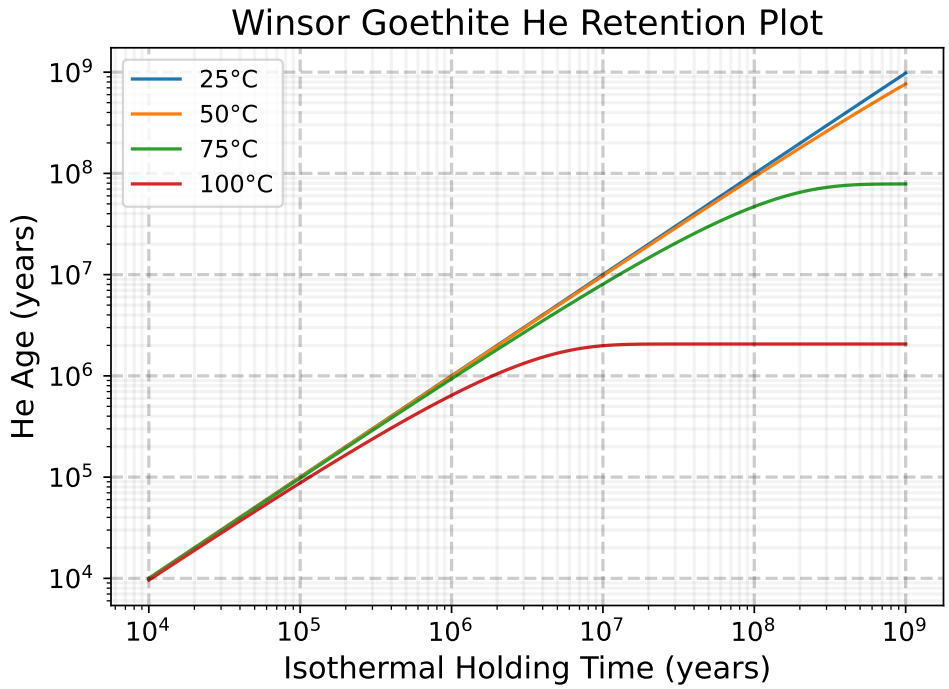


Figure 1.18: Isothermal retention plot (Wolf et al., 1998) for helium in goethite corresponding to the "Stage 1" reference line regression on the hydrothermal experiment. Helium hosted in a diffusion domain with these diffusion parameters would be nearly perfectly retentive at billion-year timescales at 25°C. At 40°C, the equilibrium age for this domain (the helium age at which radiogenic helium production and diffusive loss are equal) is 33.2 billion years, attained after 50.4 billion years.

It is difficult to use Arrhenius plots to compare many samples' helium retentivities. Farley, H. B. Monteiro, et al. (2024) report that linear regressions on apparent diffusivities calculated from helium step heat data on goethite in vacuum can be fit to yield widely variable diffusive parameters even for a single sample (e.g. activation energies ranging from 107 to 176 kJ/mol within a single experiment on YAN-0201 AC). Similarly, regressions on helium step heat data of polycrystalline materials can yield erroneous activation energies when not applied carefully (Farley, 2018). Because of the complexity of comparing Arrhenius plots of calculated apparent diffusivities for polycrystalline samples with differing heating schedules, it is advantageous to convert heating schedules to some parameter that can be compared easily between experiments. One effective way to reduce the complexity of step heat experiments to visualize helium retentivity is to "flatten" a thermal history to a dimensionless diffusive timescale: the Fourier number ($Fo = Dt/a^2$). This approach offers similar benefits to Delta plots (Farley, 2018) or r/r0 plots (Lovera et al., 1991), but is optimized to qualitatively compare the helium retentivities of polycrystalline samples for a given thermal history (expressed as diffusive time).

For a given diffusive domain, a Fourier number of $Fo \approx 1$ corresponds to the complete degassing of that domain. Complex heating schedules can be converted to Fourier numbers by assuming a reference line similarly to a Delta or r/r0 plot (Lovera et al., 1991; Farley, 2018). This procedure is especially useful for MDD (Multiple Diffusion Domain) systems, where the apparent diffusivities decrease over the course of a step heat experiment as the least retentive domains are degassed. For a single spherical diffusion domain, the fraction of helium released is a function of the Fourier number alone (Crank, 1979; Fechtig and Kalbitzer, 1966). If the activation energy of the reference Fourier number is the same as the activation energy of diffusion, then the amount of helium released in that domain then is an exact function of the reference Fourier number multiplied by a constant. Thus, a plot with reference Fourier numbers against cumulative fractional helium yields for different samples will yield the same geometries and interrelationships regardless of the intercept used to calculate the reference Fourier numbers; the values of the reference Fourier numbers will be arbitrary and only meaningful as relative diffusive time. Another implication is that different aliquots of the same sample should appear identical in this space regardless of heating schedule (see Win-06-01b in Fig. 1.20, or overlapping hematite samples in Fig. 1.19). Many such plots were generated with a variety of activation energies between 149 kJ/mol to 171 kJ/mol, but the relative positions of the degassing curves for all samples are preserved, illustrating the utility of this approach even for comparing helium retentivity between different minerals.

Fig. 1.19 compares the step heat results of Win-06-01b-HT with 11 step heat experiments of 3 samples of polycrystalline hematite and 9 step heat experiments of 2 samples of goethite in Vacuum, including Win-06-01b. In a thermal-history normalized space, it is obvious that goethite in vacuum and hematite define well-separated populations and goethite in vacuum is far less retentive of helium than hematite. Most interestingly, the most retentive sample on these plots is clearly the Win-06-01b-HT goethite under hydrothermal conditions, at least for the 14.7% of the total 3He hosted in the least retentive domains. This implies that the helium retentivity of stable goethite may even be higher than for hematite of similar size, although step heat experiments on irradiated single-crystal goethite of controlled size under hydrothermal conditions would be needed to fully assess this possibility. This plot also makes it clear that step heat results for different, unrelated goethite samples in vacuum are more similar to one another than step heat results of the same sample at different water activities.

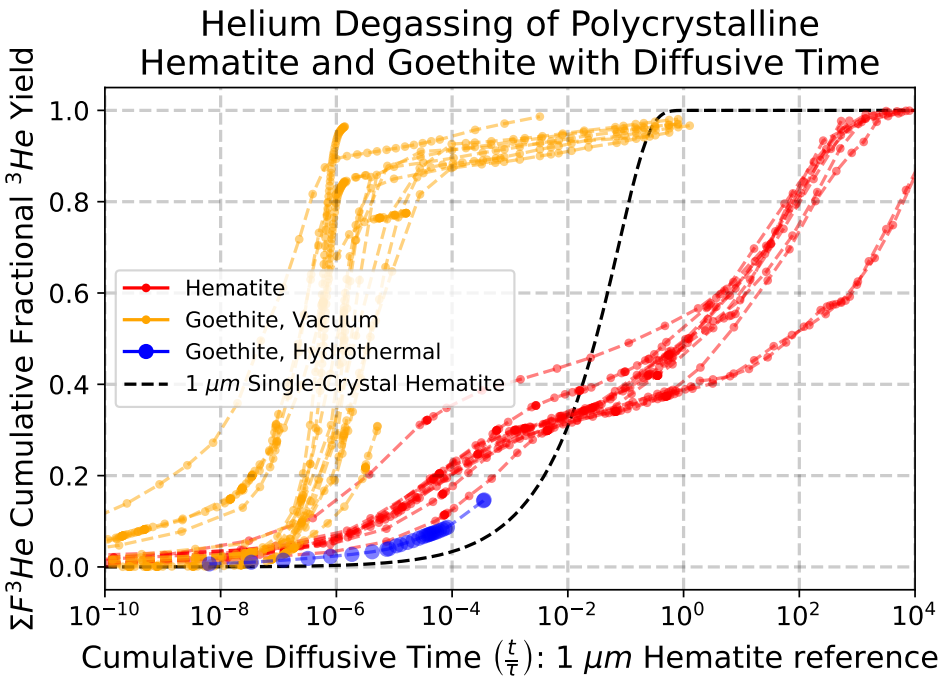


Figure 1.19: Heating schedules for polycrystalline goethite and hematite samples are flattened to reference Fourier numbers, dimensionless numbers that can be thought of as integrated thermal history, allowing for easy comparison of entire helium step heat experiments. The 3He cumulative fractional yield is plotted on the y axis as a function of Fourier number in log scale on the x axis. Curves on the left of the plot degas earlier within a given thermal history than the curves on the right. Hematite and goethite step heats in vacuum each cluster closely together, forming distinct populations. Most strikingly, the hydrothermal Win-06-01b-HT experiment is the most helium-retentive polycrystalline sample on the plot, at least until the experiment's end at a cumulative fractional yield ~15% F3He. In vacuum, all hematite samples are significantly more retentive of helium than all goethite samples, with some limited exceptions for the least retentive domains holding <5% of the total helium. Win-06-01b step heat experiments in vacuum are indistinguishable from the Capao samples in vacuum (the two leftmost goethites are Capao samples; the rest of the Capao samples closely cluster with Win-06-01b in vacuum). The differences in the curves here reflect differences in grain size distributions for hematite, and some combination of grain size and dehydration rate (due to variable water pressure in the experimental vessel and crystal aggregate size) for goethite. Vacuum step heats of 11 aliquots of 4 samples of polycrystalline hematite from the Grand Canyon, Arizona (Farley and Flowers, 2012) and the Gogebic Range, Michigan (Farley and McKeon, 2015), and 9 aliquots of Win-06-01b and Capao goethite from Minas Gerais, Brazil (H. Monteiro, 2017; Farley, H. B. Monteiro, et al., 2024) are plotted here. The reference D/a^2 values are calculated for 1µm domains of hematite ($\ln(D_0) = 0.66$; $E_a = 171$ kJ/mol: (Farley, 2018)), and the reference curve for the degassing of 1µm spherical domains of hematite with uniformly distributed 3He is calculated using the forward equations of Fechtig and Kalbitzer (1966).

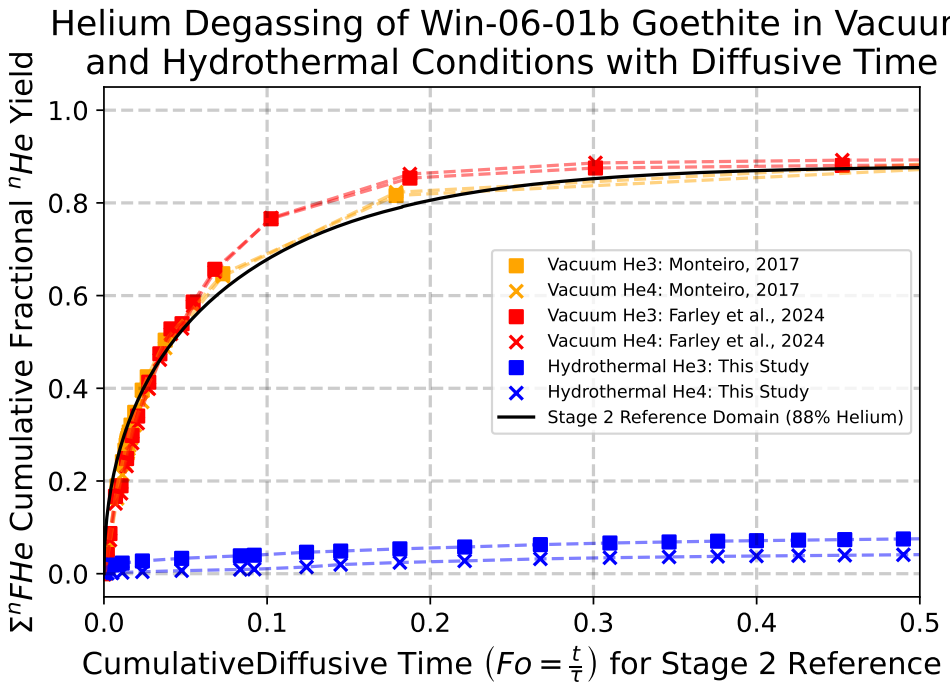


Figure 1.20: Win-06-01b step heats flattened to the Fourier number calculated from the "Stage 2" reference domain from the Win-06-01b step heat of H. Monteiro (2017). In this space, the helium retentivity contrast between the same goethite sample in vacuum and under hydrothermal conditions is striking. This plot shows that nearly 88% of helium loss for Win-06-01b in vacuum can be well-explained by a single thermally activated process following an Arrhenius law, corresponding to the "Stage 2" reference line from the experiment of H. Monteiro (2017), unlike polycrystalline hematite samples, which show a complex helium release pattern against a reference Fourier number, as expected for polycrystalline samples with complex grain size distributions. Whether or not the frequency factor D_0/a^2 for the "Stage 2" process Arrhenius law corresponds to grain size, aggregate size, or another parameter relevant to helium loss in goethite in nature is unclear. This process seems to be slightly faster for the experiment of Farley, H. B. Monteiro, et al. (2024), potentially due to lower water pressures from a liquid nitrogen trap. Both experiments of H. Monteiro (2017) and Farley, H. B. Monteiro, et al. (2024) show that helium diffusivity (and the 4/3 ratio) suddenly decreases after a cumulative fractional yield of 3He around 88%, signifying the beginning of "Stage 3" (Farley, H. B. Monteiro, et al., 2024), despite different heating schedules.

Profiling Radiogenic Helium Loss with $\frac{^4He}{^3He}$ Ratios of Win-06-01b-HT Bulk-normalized 4He/3He ratios in step heats of irradiated samples measure rounding concentration profiles by natural helium loss, if the loss mechanism in nature and in the laboratory share the same geometry (Shuster, Farley, et al., 2004; Farley, H. B. Monteiro, et al., 2024). Aside from the first experimental step, bulk-normalized 4/3 ratios for all experiments begin at vanishingly small values and steeply increase at first, with the rate of increase slowing as the experiments proceed (Fig. 1.21). This behavior is consistent with complete then partial loss of radiogenic helium in nature in the least retentive domains and is observed in all step heat experiments on irradiated goethite (Farley, H. B. Monteiro, et al., 2024). The hydrothermal experiment, however, records lower bulk-normalized 4/3 ratios than the vacuum experiments at any given $\Sigma F_{3\mathrm{He}}$ for the duration of the experiment. Similarly, the approach to the 4/3 plateau varies between the three experiments: a high bulk normalized 4/3 He plateau is most decisively represented in the Farley, H. B. Monteiro, et al. (2024) experiment, whereas the 4/3 ratios in the step heats of H. Monteiro (2017) transition more gradually before reaching a plateau later in the experiment, while the increase in 4/3 ratios is slowest for the hydrothermal experiment (Fig. 1.21). This potentially could be explained by the relative dominance of dehydration processes or their mode of spatial propagation as a function of water pressure, where the smoothest 4/3 spectra are generated by higher water pressures. The experiment in Farley, H. B. Monteiro, et al. (2024) was open to a liquid nitrogen trap during heating, which would maintain lower water pressures than H. Monteiro (2017), who in turn held Win-06-01b at much lower water pressures than the hydrothermal experiment presented in this work. While the hydrothermal experiment did not reach a plateau, it may have if the experiment was allowed to proceed further.

Interestingly, both vacuum experiments show a maximum bulk normalized 4/3 He ratio of ~ 1.1 at or before $\Sigma F_{3\mathrm{He}} = 0.88$, after which the bulk normalized 4/3 He decreases to ~0.9, suggesting that these values are independent of heating schedule and may reflect a material property of Win-06-01b.

Both vacuum experiments record a somewhat high 4/3 ratio for the first measurement that is not observed in the hydrothermal experiment (Fig. 1.21). This feature represents a very small amount of the total helium and could be related to dehydration of poorly retentive domains in the vacuum, or it may have been lost in the hydrothermal experiment while the ampoule was held at 70°C to degas entrained atmosphere before the initiation of the experiment. This small abundance of poorly retained 4He observed in the vacuum experiments could be related to physisorption of atmospheric helium (Protin et al., 2016). In any case, it represents a negligible fraction of the 4He contained in Win-06-01b.

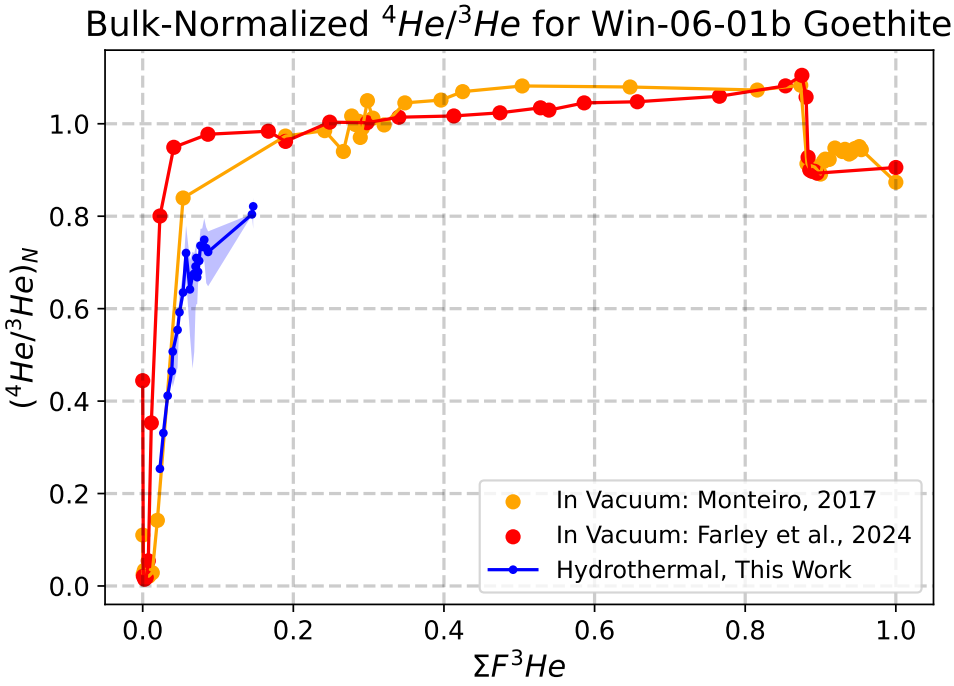


Figure 1.21: Bulk-normalized 4/3 ratios for Win-06-01b goethite step heats against cumulative fractional 3He yield. The hydrothermal experiment shows lower bulk-normalized 4/3He ratios at a given ΣF_{3He} for the entire duration of the experiment, besides the very first step. All experiments' 4/3He ratios begin at very small values and increase steeply at first, which is consistent with complete then partial loss of radiogenic helium in nature for the least retentive helium domains. The rate of increase slows between ΣF_{3He} of 0.05 to 0.2 for all experiments. The hydrothermal experiment is plotted as a shaded envelope spanning the "treatment as uncertainty" lower- and upper-bound 4/3He ratios; 10 of 34 analyses are excluded, either because the upper and lower bounds of ω cannot be computed for the early steps before λt was large enough to assume a steady-state permeation condition, or because the measured 3He signal for that step is below 0.1 pccSTP and too low-signal to give a reliable ratio.

1.7 Conclusions

Predicting the degree of helium retention and loss to guide interpretation of U-Th/He dates of natural goethite based on factors such as grain size, crystallinity, radiation damage, defect density, impurities, and aluminum substitution remains challenging. Previous efforts to understand Helium loss in natural goethite with step heating experiments in vacuum have instead queried the kinetics of dehydration (Farley, H. B. Monteiro, et al., 2024). It remains unclear how informative existing vacuum helium step heat data are with respect to helium retention and loss in natural samples. Because goethite dehydration is slowed or stopped at high water pressure, helium step heats under hydrothermal conditions are more likely to interrogate the processes controlling radiogenic helium loss in natural goethite. The method presented in this study leverages the high helium permeability of silica glass to easily conduct helium step heat experiments under hydrothermal conditions, achieving internal water pressures up to 86 bars at 300°C using standard ultra-high vacuum experimental setups (Farley, Reiners, et al., 1999).

The experiment conducted on irradiated Win-06-01b represents the first hydrothermal helium step heat experiment for goethite. Results demonstrate that apparent diffusivities under hydrothermal conditions are approximately seven natural log units lower than those observed in vacuum, suggesting that helium retention is more robust in natural environments than suggested by vacuum step heat experiments. The observed diffusivities align with a volume-diffusion loss mechanism consistent with MDD systematics, yielding an activation energy of 157.1± 5.3kJ/mol.

Further studies, including hydrothermal experiments on variable size fractions of single-crystal goethite, are necessary to confirm the mechanisms of helium migration in goethite. This methodological advancement also provides a foundation for systematically exploring the effects of factors such as grain size, crystallinity, chemical composition, and radiation damage on helium mobility in goethite.

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